CHAPTER 44
Multiple Integrals and their Applications

44.1 Evaluate the iterated integral $I = \int_2^3 \int_1^5 (x + 2y)\, dx\, dy$.

Solution $\int_1^5 (x + 2y)\, dx = (\frac{1}{2}x^2 + 2yx)\,]_1^5 = (\frac{25}{2} + 10y) - (\frac{1}{2} + 2y) = 12 + 8y$. Therefore, $I = \int_2^3 (12 + 8y)\, dy = (12y + 4y^2)\,]_2^3 = (36 + 36) - (24 + 16) = 32$.

44.2 Evaluate the iterated integral $I = \int_0^1 \int_{x^2}^{x^3} (x^2 + y^2)\, dy\, dx$.

Solution $\int_{x^2}^{x^3} (x^2 + y^2)\, dy = (x^2 y + \frac{1}{3}y^3)\,]_{x^2}^{x^3} = (x^5 + \frac{1}{3}x^9) - (x^4 + \frac{1}{3}x^6)$. Therefore, $I = \int_0^1 (x^5 + \frac{1}{3}x^9 - x^4 - \frac{1}{3}x^6)\, dx = \frac{1}{6}x^6 + \frac{1}{30}x^{10} - \frac{1}{5}x^5 - \frac{1}{21}x^7\,]_0^1 = \frac{1}{6} + \frac{1}{30} - \frac{1}{5} - \frac{1}{21} = -\frac{1}{21}$.

44.3 Evaluate the iterated integral $I = \int_{-\pi}^{\pi} \int_0^2 r \sin\theta\, dr\, d\theta$.

Solution $\int_0^2 r \sin\theta\, dr = \frac{1}{2}r^2 \sin\theta\,]_0^2 = 2\sin\theta$. Therefore, $I = \int_{-\pi}^{\pi} 2\sin\theta\, d\theta = 0$ (Problem 20.48).

44.4 Evaluate the iterated integral $I = \int_0^{\pi/2} \int_0^{\cos\theta} \rho^2 \sin\theta\, d\rho\, d\theta$.

Solution $\int_0^{\cos\theta} \rho^2 \sin\theta\, d\rho = \frac{1}{3}\rho^3 \sin\theta\,]_0^{\cos\theta} = \frac{1}{3}\cos^3\theta \sin\theta$. Hence, $I = \int_0^{\pi/2} \frac{1}{3}\cos^3\theta \sin\theta\, d\theta = -\frac{1}{12}\cos^4\theta\,]_0^{\pi/2} = -\frac{1}{12}[\cos^4(\pi/2) - \cos^4 0] = -\frac{1}{12}(0 - 1) = \frac{1}{12}$.

44.5 Evaluate the iterated integral $I = \int_0^1 \int_0^z \int_0^y (x + y + z)\, dx\, dy\, dz$.

Solution $\int_0^y (x + y + z)\, dx = \frac{1}{2}x^2 + (y + z)x\,]_0^y = \frac{1}{2}y^2 + (y + z)z$. Hence, $\int_0^z \int_0^y (x + y + z)\, dx\, dy = \int_0^z (\frac{1}{2}y^2 + yz + z^2)\, dy = \frac{1}{6}y^3 + \frac{1}{2}y^2 z + z^2 y]_0^z = \frac{1}{6}z^3 + \frac{1}{2}z^3 + z^3 = \frac{5}{3}z^3$. Therefore, $I = \int_0^1 \frac{5}{3}z^3\, dz = \frac{5}{12}z^4\,]_0^1 = \frac{5}{12}$.

44.6 Evaluate $I = \int_0^{\ln 4} \int_0^{\ln 3} e^{x+y}\, dx\, dy$.

Solution In this case the double integral may be replaced by a product: $I = (\int_0^{\ln 3} e^x\, dx)(\int_0^{\ln 4} e^y\, dy) = (3 - 1)(4 - 1) = 6$. (See Problem 44.71.)

44.7 Evaluate $\int_1^2 \int_0^y x\sqrt{y^2 - x^2}\, dx\, dy$.

Solution $\int_0^y x\sqrt{y^2 - x^2}\, dx = -\frac{1}{2} \cdot \frac{2}{3}(y^2 - x^2)^{3/2}\,]_0^y = -\frac{1}{3}(y^2 - x^2)^{3/2}\,]_0^y = -\frac{1}{3}[-(y^2)^{3/2}] = \frac{1}{3}y^3$. Therefore, $I = \int_1^2 \frac{1}{3}y^3\, dy = \frac{1}{12}y^4\,]_1^2 = \frac{1}{12}(16 - 1) = \frac{15}{12} = \frac{5}{4}$.

44.8 Evaluate $I = \int_0^1 \int_y^1 e^{x^2}\, dx\, dy$.

Solution $\int e^{x^2}\, dx$ cannot be evaluated in terms of standard functions. Therefore, we change the order of integration, using Fig. 44-1. $I = \int_0^1 \int_0^x e^{x^2}\, dy\, dx = \int_0^1 x e^{x^2}\, dx = \frac{1}{2}e^{x^2}\,]_0^1 = \frac{1}{2}(e - 1)$.

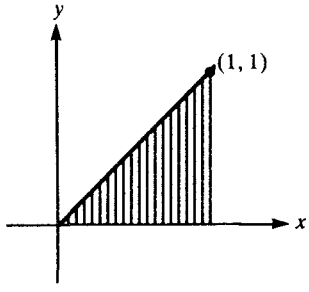


Fig. 44-1

44.9 Evaluate $\int_0^{\pi/2} \int_0^{\cos y} e^x \sin y\, dx\, dy$.

Solution $\int_0^{\cos y} e^x \sin y\, dx = (\sin y)(e^x)\,]_0^{\cos y} = (\sin y)(e^{\cos y} - 1)$. Therefore, $I = \int_0^{\pi/2} [(\sin y)e^{\cos y} - \sin y]\, dy = (-e^{\cos y} + \cos y)\,]_0^{\pi/2} = (-e^0 + 0) - (-e + 1) = e - 2$.

44.10 Evaluate $I = \int_0^1 \int_{y^4}^{y^2} \sqrt{y/x}\, dx\, dy$.

▮ $\int_{y^4}^{y^2} \sqrt{y/x}\, dx = 2\sqrt{y}\sqrt{x}\,]_{y^4}^{y^2} = 2\sqrt{y}(y - y^2) = 2(y^{3/2} - y^{5/2})$. Therefore, $I = 2\int_0^1 (y^{3/2} - y^{5/2})\, dy = 2(\frac{2}{5}y^{5/2} - \frac{2}{7}y^{7/2})\,]_0^1 = 2(\frac{2}{5} - \frac{2}{7}) = \frac{8}{35}$.

44.11 Evaluate $I = \iint_{\mathcal{R}} x\, dA$, where $\mathcal{R}$ is the region bounded by $y = x$ and $y = x^2$.

▮ The curves $y = x$ and $y = x^2$ intersect at $(0, 0)$ and $(1, 1)$, and, for $0 < x < 1$, $y = x$ is above $y = x^2$ (see Fig. 44-2). $I = \int_0^1 \int_{x^2}^{x} x\, dy\, dx = \int_0^1 xy\,]_{x^2}^{x} dx = \int_0^1 (x^2 - x^3)\, dx = (\frac{1}{3}x^3 - \frac{1}{4}x^4)\,]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.

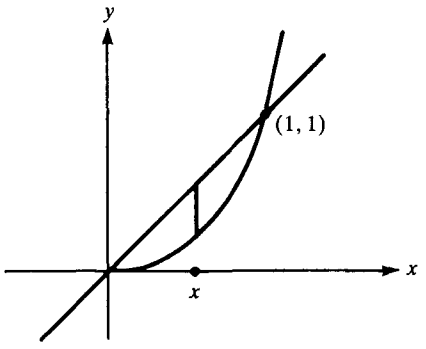


Fig. 44-2

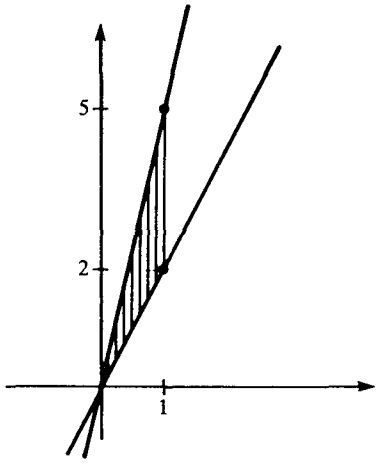


Fig. 44-3

44.12 Evaluate $I = \iint_{\mathcal{R}} y^2\, dA$, where $\mathcal{R}$ is the region bounded by $y = 2x$, $y = 5x$, and $x = 2$.

▮ The lines $y = 2x$ and $y = 5x$ intersect at the origin. For $0 < x \le 1$, the region runs from $y = 2x$ up to $y = 5x$ (Fig. 44-3). Hence, $I = \int_0^1 \int_{2x}^{5x} y^2\, dy\, dx = \int_0^1 \frac{1}{3}y^3\,]_{2x}^{5x} dx = \frac{1}{3}\int_0^1 (125x^3 - 8x^3)\, dx = \frac{1}{3}\int_0^1 117x^3\, dx = \frac{39}{4}x^4\,]_0^1 = \frac{39}{4}$.

44.13 Evaluate $I = \iint_{\mathcal{R}} (x - y)\, dA$, where $\mathcal{R}$ is the region above the x-axis bounded by $y^2 = 3x$ and $y^2 = 4 - x$ (see Fig. 44-4).

▮ It is convenient to evaluate I by means of strips parallel to the x-axis. $I = \int_0^{\sqrt{3}} \int_{y^2/3}^{4-y^2} (x - y)\, dx\, dy = \int_0^{\sqrt{3}} (\frac{1}{2}x^2 - yx)\,]_{y^2/3}^{4-y^2} dy = \int_0^{\sqrt{3}} [\frac{1}{2}(4 - y^2)^2 - y(4 - y^2)] - [\frac{1}{2}(y^2/3)^2 - y^3/3]\, dy = \int_0^{\sqrt{3}} (8 - 4y^2 + \frac{1}{2}y^4 - 4y + y^3 - \frac{1}{18}y^4 - \frac{1}{3}y^3)\, dy = \int_0^{\sqrt{3}} (8 - 4y - 4y^2 + \frac{2}{3}y^3 + \frac{4}{9}y^4)\, dy = 8y - 2y^2 - \frac{4}{3}y^3 + \frac{1}{6}y^4 + \frac{4}{45}y^5\,]_0^{\sqrt{3}} = 8\sqrt{3} - 6 - 4\sqrt{3} + \frac{3}{2} + \frac{4}{5}\sqrt{3} = \frac{24}{5}\sqrt{3} - \frac{9}{2}$.

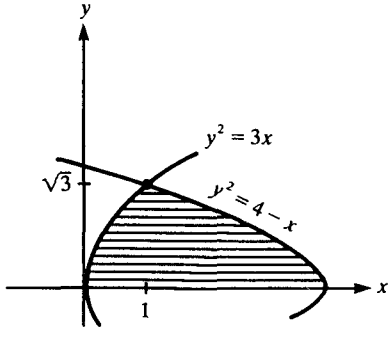


Fig. 44-4

44.14 Evaluate $I = \iint_{\mathcal{R}} \frac{1}{\sqrt{2y - y^2}}\, dA$, where $\mathcal{R}$ is the region in the first quadrant bounded by $x^2 = 4 - 2y$.

▮ The curve $x^2 = 4 - 2y$ is a parabola with vertex at $(0, 2)$ and passing through the x-axis at $x = 2$ (Fig. 44-5). Hence, $I = \int_0^2 \int_0^{\sqrt{4-2y}} \frac{1}{\sqrt{2y - y^2}}\, dx\, dy = \int_0^2 \frac{x}{\sqrt{2y - y^2}} \Big]_0^{\sqrt{4-2y}} dy = \int_0^2 \frac{\sqrt{4-2y}}{\sqrt{2y - y^2}}\, dy = \int_0^2 \frac{\sqrt{2}}{\sqrt{y}} \frac{\sqrt{2-y}}{\sqrt{2-y}}\, dy = \sqrt{2} \int_0^2 y^{-1/2}\, dy = \sqrt{2} \cdot 2y^{1/2} \Big]_0^2 = 2\sqrt{2}(\sqrt{2} - 0) = 4$. Note that, if we integrate using strips parallel to the y-axis, the integration is difficult.

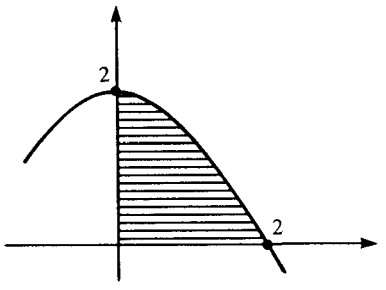


Fig. 44-5

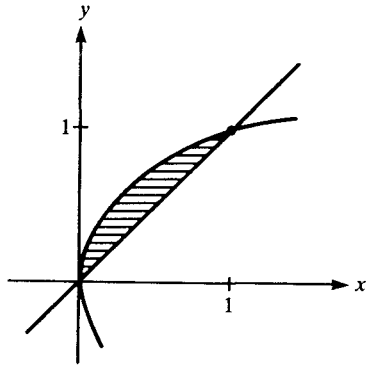


Fig. 44-6

44.15 Let $\mathcal{R}$ be the region bounded by the curve $y = \sqrt{x}$ and the line $y = x$ (Fig. 44-6). Let $f(x, y) = \frac{\sin y}{y}$ if $y \neq 0$ and $f(x, 0) = 1$. Compute $I = \iint_{\mathcal{R}} f(x, y)\, dA$.

▮ $I = \int_0^1 \int_{y^2}^{y} \frac{\sin y}{y}\, dx\, dy = \int_0^1 \frac{\sin y}{y} x \Big]_{y^2}^{y} dy = \int_0^1 (\sin y - y \sin y)\, dy$. Integration by parts yields $\int y \sin y\, dy = \sin y - y \cos y$. Hence, $I = (-\cos y + y \cos y - \sin y) \Big]_0^1 = (-\sin 1) - (-1) = 1 - \sin 1$.

44.16 Find the volume V under the plane $z = 3x + 4y$ and over the rectangle $\mathcal{R}$: $1 \le x \le 2$, $0 \le y \le 3$.

▮ $V = \iint_{\mathcal{R}} (3x + 4y)\, dA = \int_0^3 \int_1^2 (3x + 4y)\, dx\, dy = \int_0^3 (\tfrac{3}{2}x^2 + 4yx) \Big]_1^2 dy = \int_0^3 [(6 + 8y) - (\tfrac{3}{2} + 4y)]\, dy$

$= \int_0^3 (\tfrac{9}{2} + 4y)\, dy = (\tfrac{9}{2}y + 2y^2) \Big]_0^3 = \tfrac{27}{2} + 18 = \tfrac{63}{2}$.

44.17 Find the volume V in the first octant bounded by $z = y^2$, $x = 2$, and $y = 4$.

▮ $V = \iint_{\mathcal{R}} y^2\, dA = \int_0^2 \int_0^4 y^2\, dy\, dx = \int_0^2 \tfrac{1}{3}y^3 \Big]_0^4 dx = \int_0^2 \tfrac{64}{3}\, dx = \tfrac{64}{3} \cdot 2 = \tfrac{128}{3}$.

44.18 Find the volume V of the solid in the first octant bounded by $y = 0$, $z = 0$, $y = 3$, $z = x$, and $z + x = 4$ (Fig. 44-7).

▮ For given x and y, the z-value in the solid varies from $z = x$ to $z = -x + 4$. So $V = \int_0^3 \int_0^2 [(-x + 4) - x]\, dx\, dy = \int_0^3 \int_0^2 (4 - 2x)\, dx\, dy = \int_0^3 (4x - x^2) \Big]_0^2 dy = \int_0^3 (8 - 4)\, dy = 4 \int_0^3 dy = 4 \cdot 3 = 12$.

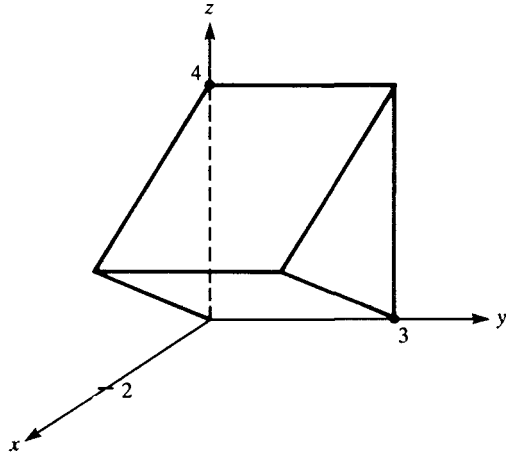


Fig. 44-7

44.19 Find the volume V of the tetrahedron bounded by the coordinate planes and the plane $z = 6 - 2x + 3y$.

▮ As shown in Fig. 44-8, the solid lies above the triangle in the xy-plane bounded by $2x + 3y = 6$ and the x and y axes. $V = \int_0^3 \int_0^{2-2x/3} (6 - 2x - 3y)\, dy\, dx = \int_0^3 6y - 2xy - \frac{3}{2}y^2 \,]_0^{2-2x/3}\, dx = \int_0^3 (2 - \frac{2}{3}x)(6 - 2x - 3 + x)\, dx = \int_0^3 \frac{2}{3}(3 - x)(3 - x)\, dx = \frac{2}{3}\int_0^3 (3 - x)^2\, dx = \frac{2}{3}(-\frac{1}{3})(3 - x)^3 \,]_0^3 = -\frac{2}{9}(-3^3) = 6$. (Check against the formula $V = \frac{1}{6}abc$.)

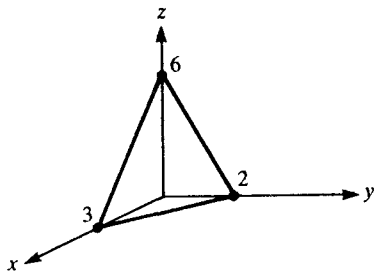


Fig. 44-8

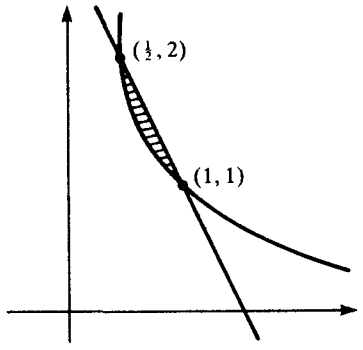


Fig. 44-9

44.20 Use a double integral to find the area of the region $\mathcal{R}$ bounded by $xy = 1$ and $2x + y = 3$.

▮ Figure 44-9 shows the region $\mathcal{R}$. $A = \iint_{\mathcal{R}} 1\, dA = \int_{1/2}^1 \int_{1/x}^{3-2x} 1\, dy\, dx = \int_{1/2}^1 (3 - 2x - 1/x)\, dx = 3x - x^2 - \ln x \,]_{1/2}^1 = (3 - 1 - 0) - (\frac{3}{2} - \frac{1}{4} - \ln \frac{1}{2}) = 2 - (\frac{5}{4} + \ln 2) = \frac{3}{4} - \ln 2$.

44.21 Find the volume V of the solid bounded by the right circular cylinder $x^2 + y^2 = 1$, the xy-plane, and the plane $x + z = 1$.

▮ As seen in Fig. 44-10, the base is the circle $x^2 + y^2 = 1$ in the xy-plane, the top is the plane $x + z = 1$. $V = \iint_{\mathcal{R}} (1 - x)\, dA = \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (1 - x)\, dy\, dx = \int_{-1}^1 (1 - x)y \,]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\, dx = \int_{-1}^1 2(1 - x)\sqrt{1 - x^2}\, dx = 2\int_{-1}^1 \sqrt{1 - x^2}\, dx - 2\int_{-1}^1 x\sqrt{1 - x^2}\, dx = 2(\pi/2) + \frac{2}{3}(1 - x^2)^{3/2} \,]_{-1}^1 = \pi + \frac{2}{3}(0) = \pi$. (Note: We know that $\int_{-1}^1 \sqrt{1 - x^2}\, dx = \pi/2$, since the integral is the area of the unit semicircle.)

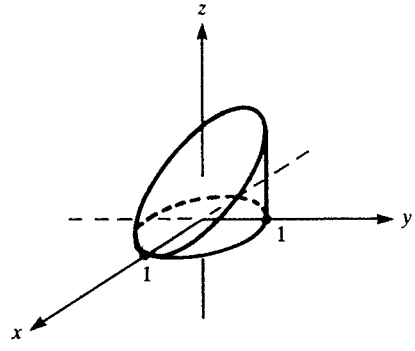


Fig. 44-10

44.22 Find the volume V of the solid bounded above by the plane $z = 3x + y + 6$, below by the xy-plane, and on the sides by $y = 0$ and $y = 4 - x^2$.

▮ Since $-2 \le x \le 2$ and $y \ge 0$, we have $z = 3x + y + 6 \ge 0$. Then $V = \int_{-2}^2 \int_0^{4-x^2} (3x + y + 6)\, dy\, dx = \int_{-2}^2 (3xy + \frac{1}{2}y^2 + 6y) \,]_0^{4-x^2}\, dx = \int_{-2}^2 [3x(4 - x^2) + \frac{1}{2}(4 - x^2)^2 + 6(4 - x^2)]\, dx = \int_{-2}^2 (32 + 12x - 10x^2 - 3x^3 + \frac{1}{2}x^4)\, dx = (32x + 6x^2 - \frac{10}{3}x^3 - \frac{3}{4}x^4 + \frac{1}{10}x^5) \,]_{-2}^2 = (64 + 24 - \frac{80}{3} - 12 + \frac{32}{10}) - (-64 + 24 + \frac{80}{3} - 12 - \frac{32}{10}) = \frac{1216}{15}$.

44.23 Find the volume of the wedge cut from the elliptical cylinder $9x^2 + 4y^2 = 36$ by the planes $z = 0$ and $z = y + 3$.

▮ On $9x^2 + 4y^2 = 36$, $-3 \le y \le 3$. Hence, $z = y + 3 \ge 0$. So the plane $z = y + 3$ will be above the plane $z = 0$ (see Fig. 44-11). Since the solid is symmetric with respect to the yz-plane, $V = 2\int_{-3}^3 \int_0^{2\sqrt{9-y^2}/3} (y + 3)\, dx\, dy = 2\int_{-3}^3 (y + 3)x \,]_0^{2\sqrt{9-y^2}/3}\, dy = 2\int_{-3}^3 (y + 3)\cdot \frac{2}{3}\sqrt{9 - y^2}\, dy = \frac{4}{3}\int_{-3}^3 y\sqrt{9 - y^2}\, dy + 4\int_{-3}^3 \sqrt{9 - y^2}\, dy = 0 + 4 \cdot \frac{1}{2}(9\pi) = 18\pi$. [The integral $\int_{-3}^3 \sqrt{9 - y^2}\, dy$ represents the area of the upper semicircle of the circle $x^2 + y^2 = 9$. Hence, it is equal to $\frac{1}{2} \cdot \pi(3)^2 = \frac{1}{2} \cdot 9\pi$.]

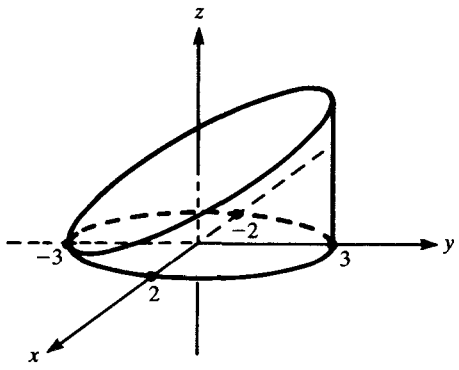


Fig. 44-11

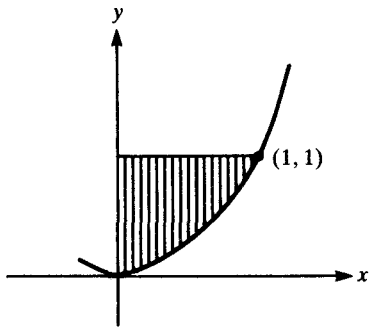


Fig. 44-12

44.24 Express the integral $I = \int_0^1 \int_0^{\sqrt{y}} f(x, y)\, dx\, dy$ as an integral with the order of integration reversed.

▮ In the region of integration, the x-values for $0 \le y \le 1$ range from 0 to $\sqrt{y}$. Hence, the bounding curve is $x = \sqrt{y}$, or $y = x^2$. Thus (see Fig. 44-12), $I = \int_0^1 \int_{x^2}^1 f(x, y)\, dy\, dx$.

44.25 Express the integral $I = \int_0^4 \int_{x/2}^2 f(x, y)\, dy\, dx$ as an integral with the order of integration reversed.

▮ For $0 \le x \le 4$, the region of integration runs from $x/2$ to 2. Hence, the region of integration is the triangle indicated in Fig. 44-13. So, if we use strips parallel to the x-axis, $I = \int_0^2 \int_0^{2y} f(x, y)\, dx\, dy$.

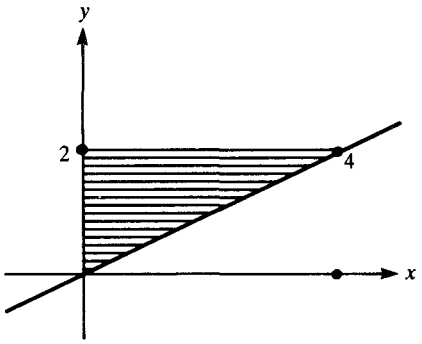


Fig. 44-13

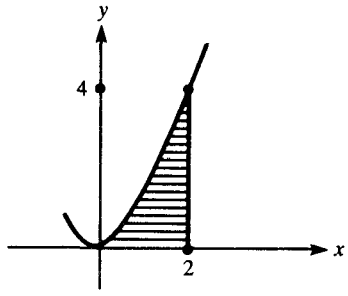


Fig. 44-14

44.26 Express $I = \int_0^2 \int_0^{x^2} f(x, y)\, dy\, dx$ as a double integral with the order of integration reversed.

▮ The region of integration is bounded by $y = 0$, $x = 2$, and $y = x^2$ (Fig. 44-14). $I = \int_0^4 \int_{\sqrt{y}}^2 f(x, y)\, dx\, dy$.

44.27 Express $I = \int_0^{\pi/2} \int_0^{\cos x} x^2\, dy\, dx$ as double integral with the order of integration reversed and compute its value.

▮ The region of integration is bounded by $y = \cos x$, $y = 0$, and $x = 0$ (Fig. 44-15). So $I = \int_0^1 \int_0^{\cos^{-1} y} x^2\, dx\, dy$. The original form is easier to calculate. $I = \int_0^{\pi/2} x^2 y \,]_0^{\cos x}\, dx = \int_0^{\pi/2} x^2 \cos x\, dx$. Two integrations by parts yields $\int x^2 \cos x\, dx = x^2 \sin x + 2x \cos x - 2 \sin x$. Hence, $I = (x^2 \sin x + 2x \cos x - 2 \sin x) \,]_0^{\pi/2} = \pi^2/4 - 2$.

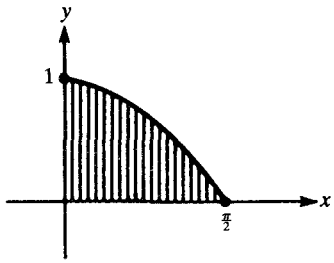


Fig. 44-15

44.28 Find $I = \iint_{\mathcal{R}} e^{x^3}\,dA$, where $\mathcal{R}$ is the region bounded by $y = x^2$, $x = 3$, and $y = 0$.

I Use strips parallel to the y-axis (see Fig. 44-16). $I = \int_0^3 \int_0^{x^2} e^{x^3}\,dy\,dx = \int_0^3 e^{x^3} y\,]_0^{x^2}\,dx = \int_0^3 e^{x^3} x^2\,dx = \frac{1}{3} e^{x^3}\,]_0^3 = \frac{1}{3}(e^{27} - e^0) = \frac{1}{3}(e^{27} - 1)$. Note that the integral with the variables in reverse order would have been impossible to calculate.

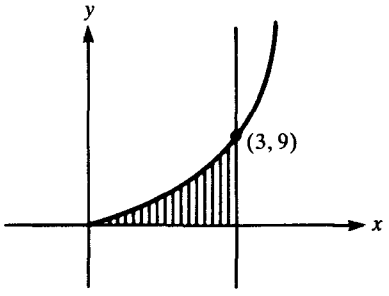


Fig. 44-16

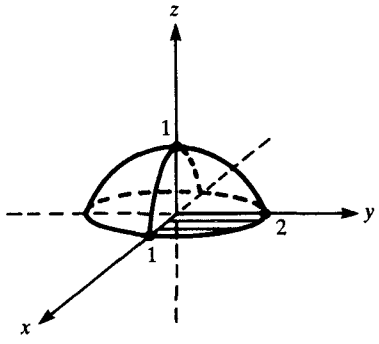


Fig. 44-17

44.29 Find the volume cut from $4x^2 + y^2 + 4z = 4$ by the plane $z = 0$.

I The elliptical paraboloid $4x^2 + y^2 + 4z = 4$ has its vertex at $(0, 0, 1)$ and opens downward. It cuts the xy-plane in an ellipse, $4x^2 + y^2 = 4$, which is the boundary of the base $\mathcal{R}$ of the solid whose volume is to be computed (see Fig. 44-17). Because of symmetry, we need to integrate only over the first-quadrant portion of $\mathcal{R}$ and then multiply by 4. $V = 4 \cdot \frac{1}{4} \int_0^1 \int_0^{\sqrt{4-4x^2}} (4 - 4x^2 - y^2)\,dy\,dx = \int_0^1 [(4 - 4x^2)y - \frac{1}{3}y^3]\,]_0^{\sqrt{4-4x^2}}\,dx = \int_0^1 \sqrt{4 - 4x^2}[4 - 4x^2 - \frac{1}{3}(4 - 4x^2)]\,dx = \int_0^1 \frac{2}{3}(4 - 4x^2)^{3/2}\,dx = \frac{2}{3}\int_0^1 8(1 - x^2)^{3/2}\,dx = \frac{16}{3}\int_0^1 (1 - x^2)^{3/2}\,dx$. Let $x = \sin\theta$, $dx = \cos\theta\,d\theta$. Then $V = \frac{16}{3}\int_0^{\pi/2} \cos^4\theta\,d\theta = \dfrac{16}{3}\displaystyle\int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2}\right)^2 d\theta = \frac{4}{3}\int_0^{\pi/2} (1 + 2\cos 2\theta + \cos^2 2\theta)\,d\theta = \dfrac{4}{3}\displaystyle\int_0^{\pi/2} \left(1 + 2\cos 2\theta + \frac{1 + \cos 4\theta}{2}\right) d\theta = \frac{4}{3}(\frac{3}{2}\theta + \sin 2\theta + \frac{1}{8}\sin 4\theta)\,]_0^{\pi/2} = \frac{4}{3}(3\pi/4) = \pi$.

44.30 Find the volume in the first octant bounded by $x^2 + z = 64$, $3x + 4y = 24$, $x = 0$, $y = 0$, and $z = 0$.

I See Fig. 44-18. The roof of the solid is given by $z = 64 - x^2$. The base $\mathcal{R}$ is the triangle in the first quadrant of the xy-plane bounded by the line $3x + 4y = 24$ and the coordinate axes. Hence, $V = \iint_{\mathcal{R}} (64 - x^2)\,dA = \int_0^8 \int_0^{(24-3x)/4} (64 - x^2)\,dy\,dx = \int_0^8 (64 - x^2)\,]_0^{(24-3x)/4}\,dx = \int_0^8 (64 - x^2) \cdot \frac{3}{4}(8 - x)\,dx = \frac{3}{4}\int_0^8 (512 - 64x - 8x^2 + x^3)\,dx = \frac{3}{4}(512x - 32x^2 - \frac{8}{3}x^3 + \frac{1}{4}x^4)\,]_0^8 = \frac{3}{4}(2^{12} - 2^{11} - \frac{1}{3} \cdot 2^{12} + \frac{1}{4} \cdot 2^{12}) = 3 \cdot 2^{10}(1 - \frac{1}{2} - \frac{1}{3} + \frac{1}{4}) = 3 \cdot 2^{10} \cdot \frac{5}{12} = 2^8 \cdot 5 = 1280$.

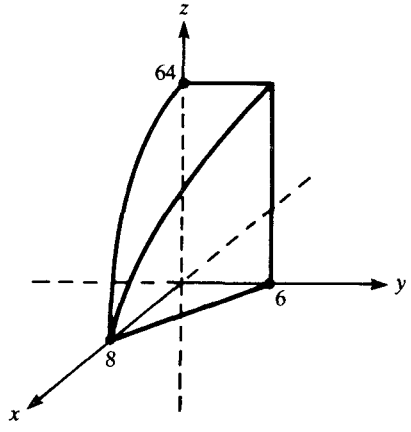


Fig. 44-18

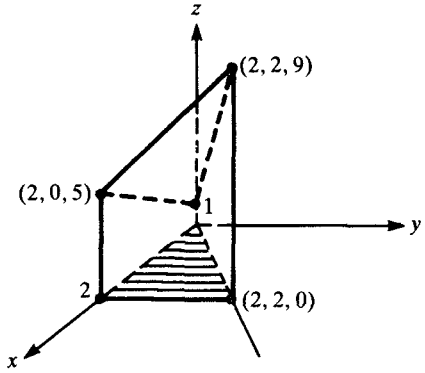


Fig. 44-19

44.31 Find the volume in the first octant bounded by $2x + 2y - z + 1 = 0$, $y = x$, and $x = 2$.

I See Fig. 44-19. $V = \iint_{\mathcal{R}} (2x + 2y + 1)\,dA = \int_0^2 \int_0^x (2x + 2y + 1)\,dy\,dx = \int_0^2 (2xy + y^2 + y)\,]_0^x\,dx = \int_0^2 (3x^2 + x)\,dx = (x^3 + \frac{1}{2}x^2)\,]_0^2 = 8 + 2 = 10$.

44.32 Find the volume of a wedge cut from the cylinder $4x^2 + y^2 = a^2$ by the planes $z = 0$ and $z = my$.

▮ The base is the semidisk $\mathcal{R}$ bounded by the ellipse $4x^2 + y^2 = a^2$, $y \geq 0$. Because of symmetry, we need only double the first-octant volume. Thus, $V = 2\int_0^{a/2}\int_0^{\sqrt{a^2-4x^2}} my\,dy\,dx = m\int_0^{a/2} y^2\,]_0^{\sqrt{a^2-4x^2}}\,dx = m\int_0^{a/2}(a^2 - 4x^2)\,dx = m(a^2x - \frac{4}{3}x^3)\,]_0^{a/2} = m\cdot(a/2)[a^2 - \frac{4}{3}(a^2/4)] = (ma/2)(\frac{2}{3}a^2) = (m/3)a^3$.

44.33 Find $I = \iint_{\mathcal{R}} \sin\theta\,dA$, where $\mathcal{R}$ is the region outside the circle $r = 1$ and inside the cardioid $r = 1 + \cos\theta$ (see Fig. 44-20).

▮ For polar coordinates, recall that the factor r is introduced into the integrand via $dA = r\,dr\,d\theta$. By symmetry, we can restrict the integration to the first quadrant and double the result. $I = 2\int_0^{\pi/2}\int_1^{1+\cos\theta}(\sin\theta)\,r\,dr\,d\theta = 2\int_0^{\pi/2}\frac{1}{2}(\sin\theta)r^2\,]_1^{1+\cos\theta}\,d\theta = \int_0^{\pi/2}[(1+\cos\theta)^2\sin\theta - \sin\theta]\,d\theta = [-(1+\cos\theta)^3/3 + \cos\theta]\,]_0^{\pi/2} = -\frac{1}{3} - (-\frac{8}{3} + 1) = \frac{4}{3}$.

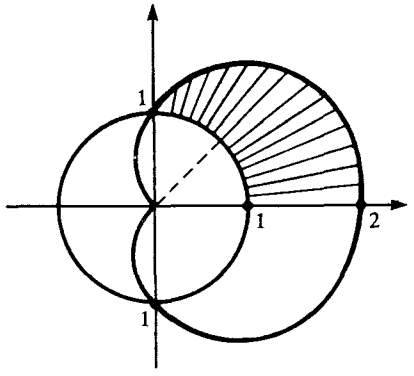


Fig. 44-20

44.34 Use cylindrical coordinates to calculate the volume of a sphere of radius a.

▮ In cylindrical coordinates, the sphere with center $(0,0,0)$ is $r^2 + z^2 = a^2$. Calculate the volume in the first octant and multiply it by 8. The base is the quarter disk $0 \leq r \leq a$, $0 \leq \theta \leq \pi/2$. $V = 8\int_0^{\pi/2}\int_0^a \sqrt{a^2 - r^2}\,r\,dr\,d\theta = 8\int_0^{\pi/2} -\frac{1}{2}\cdot\frac{2}{3}(a^2 - r^2)^{3/2}\,]_0^a\,d\theta = 8\int_0^{\pi/2}\frac{1}{3}a^3\,d\theta = (8a^3/3)\int_0^{\pi/2}d\theta = (8a^3/3)(\pi/2) = \frac{4}{3}\pi a^3$, the standard formula.

44.35 Use polar coordinates to evaluate $I = \int_0^1\int_0^{\sqrt{1-x^2}}(x^2 + y^2)^{5/2}\,dy\,dx$.

▮ The region of integration is the part of the unit disk in the first quadrant: $0 \leq \theta \leq \pi/2$, $0 \leq r \leq 1$. Hence, $I = \int_0^{\pi/2}\int_0^1 r^5\cdot r\,dr\,d\theta = \int_0^{\pi/2}\int_0^1 r^6\,dr\,d\theta = \int_0^{\pi/2}\frac{1}{7}r^7\,]_0^1\,d\theta = \frac{1}{7}\int_0^{\pi/2}d\theta = \frac{1}{7}(\pi/2) = \pi/14$.

44.36 Find the area of the region enclosed by the cardioid $r = 1 + \cos\theta$.

▮ $A = \iint_{\mathcal{R}} 1\,dA = \int_0^{2\pi}\int_0^{1+\cos\theta} r\,dr\,d\theta = \int_0^{2\pi}\frac{1}{2}r^2\,]_0^{1+\cos\theta}\,d\theta = \frac{1}{2}\int_0^{2\pi}(1 + 2\cos\theta + \cos^2\theta)\,d\theta = \frac{1}{2}\int_0^{2\pi}\left(1 + 2\cos\theta + \frac{1+\cos 2\theta}{2}\right)d\theta = \frac{1}{2}(\frac{3}{2}\theta + 2\sin\theta + \frac{1}{2}\sin 2\theta)\,]_0^{2\pi} = \frac{1}{2}(\frac{3}{2}\cdot 2\pi) = \frac{3\pi}{2}$.

44.37 Use polar coordinates to find the area of the region inside the circle $x^2 + y^2 = 9$ and to the right of the line $x = \frac{3}{2}$.

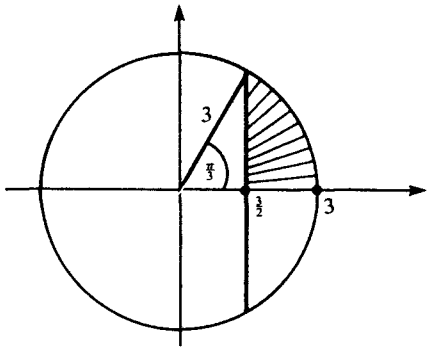


Fig. 44-21

▮ It suffices to double the area in the first quadrant. $x^2 + y^2 = 9$ becomes $r = 3$ in polar coordinates, and $x = \frac{3}{2}$ is equivalent to $r\cos\theta = \frac{3}{2}$. From Fig. 44-21, $0 \le \theta \le \pi/3$. So the area is $2\int_0^{\pi/3}\int_{3/(2\cos\theta)}^{3} r\,dr\,d\theta = \int_0^{\pi/3} r^2\,]_{(3\sec\theta)/2}^{3}\,d\theta = \int_0^{\pi/3}(9 - \frac{9}{4}\sec^2\theta)\,d\theta = (9\theta - \frac{9}{4}\tan\theta)\,]_0^{\pi/3} = 3\pi - (9\sqrt{3}/8)$.

44.38 Describe the planar region $\mathcal{R}$ whose area is given by the iterated integral $I = \int_{\pi}^{2\pi}\int_{1}^{1-\sin\theta} r\,dr\,d\theta$.

▮ $r = 1 - \sin\theta$ is a cardioid, and $r = 1$ is the unit circle. Between $\theta = \pi$ and $\theta = 2\pi$, $1 < 1 - \sin\theta$ and the cardioid is outside the circle. Therefore, $\mathcal{R}$ is the region outside the unit circle and inside the cardioid (Fig. 44-22).

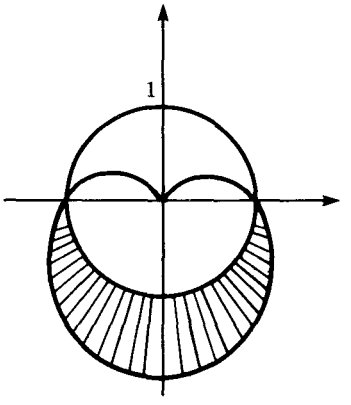


Fig. 44-22

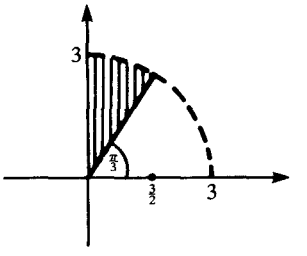


Fig. 44-23

44.39 Evaluate the integral $I = \int_0^{3/2}\int_{\sqrt{3}x}^{\sqrt{9-x^2}} 2xy\,dy\,dx$ using (*a*) rectangular coordinates and (*b*) polar coordinates.

▮ (*a*) $I = \int_0^{3/2} xy^2\,]_{\sqrt{3}x}^{\sqrt{9-x^2}}\,dx = \int_0^{3/2}[x(9-x^2) - 3x^3]\,dx = \int_0^{3/2}(9x - 4x^3)\,dx = (\frac{9}{2}x^2 - x^4)\,]_0^{3/2} = \frac{9}{4}(\frac{9}{2} - \frac{9}{4}) = (\frac{9}{4})^2 = \frac{81}{16}$. (*b*) As indicated in Fig. 44-23, the region of integration lies under the semicircle $y = \sqrt{9 - x^2}$ (or $r = 3$) and above the line $y = \sqrt{3}\,x$ (or $\theta = \pi/3$). Hence, $I = 2\int_{\pi/3}^{\pi/2}\int_0^3 r\cos\theta \cdot r\cos\theta \cdot r\,dr\,d\theta = 2\int_{\pi/3}^{\pi/2}\int_0^3 r^3\cos\theta\sin\theta\,dr\,d\theta = \frac{1}{2}\int_{\pi/3}^{\pi/2} r^4\cos\theta\sin\theta\,]_0^3\,d\theta = \frac{81}{2}\int_{\pi/3}^{\pi/2}\cos\theta\sin\theta\,d\theta = \frac{81}{2}\cdot\frac{1}{2}\sin^2\theta\,]_{\pi/3}^{\pi/2} = \frac{81}{4}(1 - \frac{3}{4}) = \frac{81}{16}$.

44.40 Find the volume of the solid cut out from the sphere $x^2 + y^2 + z^2 \le 4$ by the cylinder $x^2 + y^2 = 1$ (see Fig. 44-24).

▮ It suffices to multiply by 8 the volume of the solid in the first octant. Use cylindrical coordinates. The sphere is $r^2 + z^2 = 4$ and the cylinder is $r = 1$. Thus, we have $V = 8\int_0^{\pi/2}\int_0^1 \sqrt{4 - r^2}\,r\,dr\,d\theta = 8\int_0^{\pi/2}[-\frac{1}{2}\cdot\frac{2}{3}(4 - r^2)^{3/2}]\,]_0^1\,d\theta = -\frac{8}{3}\int_0^{\pi/2}[(3)^{3/2} - 8]\,d\theta = \frac{8}{3}(8 - 3\sqrt{3})\int_0^{\pi/2}d\theta = \frac{4\pi}{3}(8 - 3\sqrt{3})$.

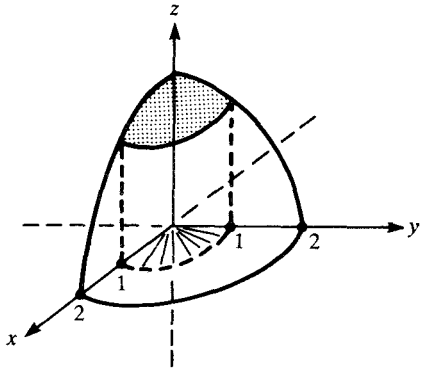


Fig. 44-24

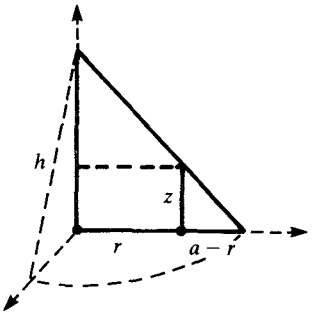


Fig. 44-25

44.41 Use integration in cylindrical coordinates to find the volume of a right circular cone of radius a and height h.

▮ The base is the disk of radius a, given by $r \le a$. For given r, the corresponding value of z on the cone is determined by $z/(a - r) = h/a$ (obtained by similar triangles; see Fig. 44-25.) Then $V = \int_0^{2\pi}\int_0^a (h/a)(a - r)r\,dr\,d\theta = \int_0^{2\pi}[\frac{1}{2}hr^2 - (h/3a)r^3\,]_0^a\,d\theta = \int_0^{2\pi}(\frac{1}{2}ha^2 - \frac{1}{3}ha^2)\,d\theta = \frac{1}{6}ha^2\int_0^{2\pi}d\theta = \frac{1}{6}ha^2(2\pi) = \frac{1}{3}\pi a^2 h$, the standard formula.

44.42 Find the average distance from points in the unit disk to a fixed point on the boundary.

▮ For the unit circle $r = 2\sin\theta$, with the pole as the fixed point (Fig. 44-26), the distance of an interior point to the pole is r. Thus, $\bar{r} = \dfrac{1}{\text{area of } \mathcal{R}} \cdot \iint_{\mathcal{R}} r\,dA = \dfrac{1}{\pi}\int_0^{\pi}\int_0^{2\sin\theta} r \cdot r\,dr\,d\theta = \dfrac{1}{\pi}\int_0^{\pi} \dfrac{1}{3} r^3 \Big]_0^{2\sin\theta} d\theta =$ $\dfrac{8}{3\pi}\int_0^{\pi} \sin^3\theta\,d\theta = \dfrac{8}{3\pi}\int_0^{\pi} (\sin\theta - \cos^2\theta\sin\theta)\,d\theta = \dfrac{8}{3\pi}\left(-\cos\theta + \tfrac{1}{3}\cos^3\theta\right)\Big]_0^{\pi} = \dfrac{8}{3\pi}[(1 - \tfrac{1}{3}) - (-1 + \tfrac{1}{3})] =$ $\dfrac{8}{3\pi}\left(\dfrac{4}{3}\right) = \dfrac{32}{9\pi}$.

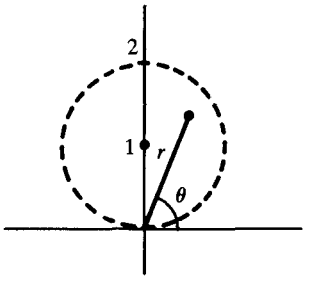


Fig. 44-26

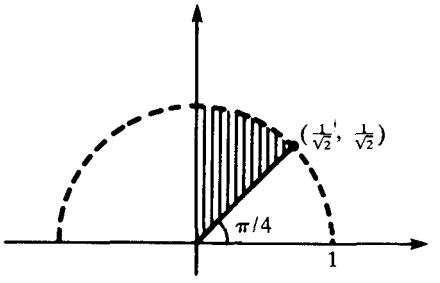


Fig. 44-27

44.43 Use polar coordinates to evaluate $I = \int_0^{1/\sqrt{2}} \int_x^{\sqrt{1-x^2}} \sqrt{x^2 + y^2}\,dy\,dx$.

▮ The region of integration is indicated in Fig. 44-27. $I = \int_{\pi/4}^{\pi/2}\int_0^1 r \cdot r\,dr\,d\theta = (\int_0^1 r^2\,dr)(\int_{\pi/4}^{\pi/2} d\theta) = \left(\dfrac{1}{3}\right)\left(\dfrac{\pi}{4}\right) = \dfrac{\pi}{12}$.

44.44 Use polar coordinates to evaluate $I = \int_0^2 \int_0^{\sqrt{2x-x^2}} x\,dy\,dx$.

▮ $y = \sqrt{2x - x^2}$ is equivalent to $y^2 = 2x - x^2$, $y \geq 0$, or the upper half of the circle $(x-1)^2 + y^2 = 1$ (see Fig. 44-28). In polar coordinates, $x^2 + y^2 = 2x$ is equivalent to $r^2 = 2r\cos\theta$, or $r = 2\cos\theta$. Thus, $I = \int_0^{\pi/2}\int_0^{2\cos\theta} (r\cos\theta) r\,dr\,d\theta = \int_0^{\pi/2}\int_0^{2\cos\theta} r^2\cos\theta\,dr\,d\theta = \int_0^{\pi/2} \tfrac{1}{3}r^3\cos\theta\,]_0^{2\cos\theta}\,d\theta = \tfrac{8}{3}\int_0^{\pi/2}\cos^4\theta\,d\theta = \dfrac{\pi}{2}$ (by Problem 44.29).

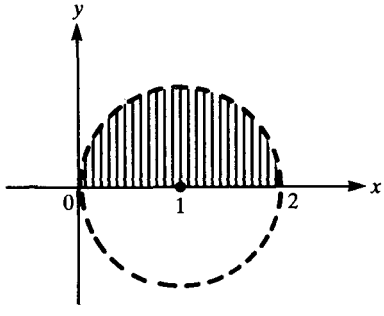


Fig. 44-28

44.45 If the depth of water provided by a water sprinkler in a given unit of time is 2^{-r} feet at a distance of r feet from the sprinkler, find the total volume of water within a distance of a feet from the sprinkler after one unit of time.

▮ $$V = \int_0^{2\pi} d\theta \int_0^a e^{-r\ln 2}\,r\,dr = 2\pi\left[\frac{e^{-r\ln 2}}{(\ln 2)^2}(-r\ln 2 - 1)\right]_0^a = \frac{2\pi}{(\ln 2)^2}[1 - e^{-a\ln 2}(1 + a\ln 2)]$$

$$= \frac{2\pi}{(\ln 2)^2}\left(1 - \frac{1 + a\ln 2}{2^a}\right).$$

44.46 Evaluate $I = \int_{\sqrt{2}/2}^1 \int_{\sqrt{1-x^2}}^x \dfrac{1}{\sqrt{x^2 + y^2}}\,dy\,dx$.

▮ The region of integration (Fig. 44-29) consists of all points in the first quadrant above the circle $x^2 + y^2 = 1$ and under the line $y = x$. Transform to polar coordinates, noting that $x = 1$ is equivalent to $r = \sec\theta$. $I = \int_0^{\pi/4}\int_1^{\sec\theta} \dfrac{1}{r} \cdot r\,dr\,d\theta = \int_0^{\pi/4}\int_1^{\sec\theta} dr\,d\theta = \int_0^{\pi/4} (\sec\theta - 1)\,d\theta = (\ln|\sec\theta + \tan\theta| - \theta)\,]_0^{\pi/4} = \ln(\sqrt{2} + 1) - \pi/4$.

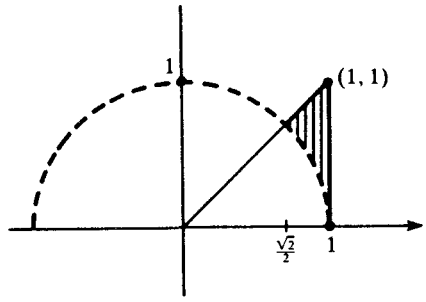


Fig. 44-29

44.47 Show that $\int_0^\infty e^{-x^2}\,dx = \dfrac{\sqrt{\pi}}{2}$.

▮ Let $c = \int_0^\infty e^{-x^2}\,dx$. Then $c = \int_0^\infty e^{-y^2}\,dy$. Hence, $c^2 = \int_0^\infty e^{-x^2}\,dx \cdot \int_0^\infty e^{-y^2}\,dy = \int_0^\infty e^{-y^2}(\int_0^\infty e^{-x^2}\,dx)\,dy = \int_0^\infty \int_0^\infty e^{-x^2}e^{-y^2}\,dx\,dy = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy$. The region of integration is the entire first quadrant. Change to polar coordinates. $c^2 = \int_0^\infty \int_0^{\pi/2} e^{-r^2}\,r\,d\theta\,dr = \int_0^\infty e^{-r^2}\cdot r\theta\,]_0^{\pi/2}\,dr = \dfrac{\pi}{2}\int_0^\infty e^{-r^2}\,r\,dr = (\pi/2)(-\tfrac{1}{2})e^{-r^2}\,]_0^\infty = -(\pi/4)(\lim_{r\to+\infty} e^{-r^2} - e^0) = -(\pi/4)(0-1) = \pi/4$. Since $c^2 = \pi/4$, $c = \sqrt{\pi}/2$. (The rather loose reasoning in this computation can be made rigorous.)

44.48 Evaluate $I = \int_0^\infty \dfrac{e^{-x}}{\sqrt{x}}\,dx$.

▮ Let $u = \sqrt{x}$, $x = u^2$, $dx = 2u\,du$. Then $I = \int_0^\infty \dfrac{e^{-u^2}}{u}\cdot 2u\,du = 2\int_0^\infty e^{-u^2}\,du = 2\cdot\dfrac{\sqrt{\pi}}{2} = \sqrt{\pi}$, by Problem 44.47.

44.49 Evaluate $I = \int_0^\infty x^2e^{-x^2}\,dx$.

▮ Consider $I_w = \int_0^w x^2e^{-x^2}\,dx$. Use integration by parts. Let $u = x$, $dv = xe^{-x^2}\,dx$, $du = dx$, $v = -\frac{1}{2}e^{-x^2}$. Then $I_w = -\frac{1}{2}xe^{-x^2}\,]_0^w + \frac{1}{2}\int_0^w e^{-x^2}\,dx$. Hence, $I = \lim_{w\to+\infty} I_w = \lim_{w\to+\infty}(-\frac{1}{2}we^{-w^2}) + \frac{1}{2}\int_0^\infty e^{-x^2}\,dx = 0 + \frac{1}{2}\sqrt{\pi}/2 = \sqrt{\pi}/4$, by Problem 44.47.

44.50 Use spherical coordinates to find the volume of a sphere of radius a.

▮ In spherical coordinates a sphere of radius a is characterized by $0\le\rho\le a$, $0\le\theta\le 2\pi$, $0\le\phi\le\pi$. Recall that the volume element is given by $dV = \rho^2\sin\phi\,d\rho\,d\theta\,d\phi$. $V = \int_0^\pi\int_0^{2\pi}\int_0^a \rho^2\sin\phi\,d\rho\,d\theta\,d\phi = \int_0^\pi\int_0^{2\pi}\frac{1}{3}\rho^3\sin\phi\,]_0^a\,d\theta\,d\phi = \int_0^\pi\int_0^{2\pi}\frac{1}{3}a^3\sin\phi\,d\theta\,d\phi = \int_0^\pi\frac{1}{3}a^3\sin\phi\cdot\theta\,]_0^{2\pi}\,d\phi = \int_0^\pi\frac{1}{3}a^3\sin\phi\cdot 2\pi\,d\phi = \frac{2}{3}\pi a^3\int_0^\pi\sin\phi\,d\phi = -\frac{2}{3}\pi a^3\cos\phi\,]_0^\pi = -\frac{2}{3}\pi a^3(-1-1) = \frac{4}{3}\pi a^3$.

44.51 Use spherical coordinates to find the volume of a right circular cone of height h and radius of base b.

▮ For the orientation shown in Fig. 44-30, the points of the cone satisfy $0\le\theta\le 2\pi$, $0\le\phi\le\tan^{-1}(b/h)$, $0\le\rho\le h\sec\phi$. Thus, $V = \int_0^{2\pi}\int_0^{\tan^{-1}(b/h)}\int_0^{h\sec\phi}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \int_0^{2\pi}\int_0^{\tan^{-1}(b/h)}\frac{1}{3}\rho^3\sin\phi\,]_0^{h\sec\phi}\,d\phi\,d\theta = \int_0^{2\pi}\int_0^{\tan^{-1}(b/h)}\frac{1}{3}\,h^3\,\frac{\sin\phi}{\cos^3\phi}\,d\phi\,d\theta = \frac{1}{3}h^3\int_0^{2\pi}\frac{1}{2\cos^2\phi}\,]_0^{\tan^{-1}(b/h)}\,d\theta = \frac{1}{3}h^3\int_0^{2\pi}\frac{1}{2}\left(\frac{s^2}{h^2}-1\right)d\theta = \frac{h^3}{6h^2}\int_0^{2\pi}(s^2-h^2)\,d\theta = \frac{hb^2}{6}\int_0^{2\pi}d\theta = \frac{hb^2}{6}\cdot 2\pi = \frac{1}{3}\pi b^2h$.

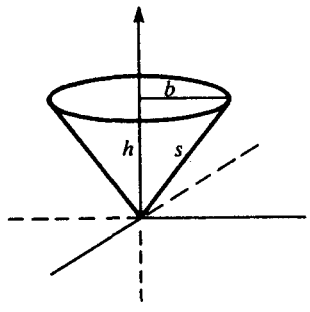


Fig. 44-30

44.52 Find the average distance $\bar{\rho}$ from the center of a ball $\mathcal{B}$ of radius a to all other points of the ball.

▮ $\bar{\rho} = \dfrac{1}{\text{volume } V} \cdot \iiint_{\mathcal{B}} \rho \, dV = \dfrac{1}{\frac{4}{3}\pi a^3} \int_0^{2\pi} \int_0^{\pi} \int_0^{a} \rho \cdot \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \dfrac{3}{4\pi a^3} \int_0^{2\pi} \int_0^{\pi} \dfrac{1}{4} \rho^4 \sin\phi \Big]_0^{a} d\phi \, d\theta =$
$\dfrac{3}{4\pi a^3} \int_0^{2\pi} \int_0^{\pi} \dfrac{1}{4} a^4 \sin\phi \, d\phi \, d\theta = \dfrac{3a}{16\pi} \int_0^{2\pi} -\cos\phi \Big]_0^{\pi} d\theta = \dfrac{3a}{16\pi} \int_0^{2\pi} -(-1-1) \, d\theta = \dfrac{3a}{8\pi} \int_0^{2\pi} d\theta = \dfrac{3a}{8\pi} \cdot 2\pi = \dfrac{3}{4}a.$

44.53 Find the area S of the part of the plane $x + 2y + z = 4$ which lies inside the cylinder $x^2 + y^2 = 1$.

▮ Recall (Fig. 44-31) the relation $dA = dS \cos\theta = dS \dfrac{1}{\sqrt{1 + z_x^2 + z_y^2}}$ between an element of area dS of a surface $z = f(x, y)$ and its projection dA in the xy-plane. Thus, the formula for S is $\iint_{\mathcal{R}} \sqrt{1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2} \, dA$, where, here, $\mathcal{R}$ is the disk $x^2 + y^2 \le 1$. $1 + \dfrac{\partial z}{\partial x} = 0$, $\dfrac{\partial z}{\partial x} = -1$. $2 + \dfrac{\partial z}{\partial y} = 0$, $\dfrac{\partial z}{\partial y} = -2$. Hence, $S = \iint_{\mathcal{R}} \sqrt{6} \, dA = \sqrt{6}(\text{area } \mathcal{R}) = \sqrt{6}\,\pi$.

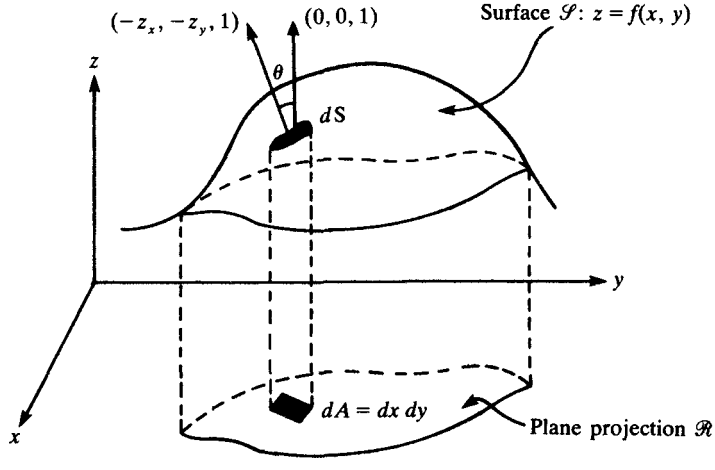


Fig. 44-31

44.54 Find the surface area S of the part of the sphere $x^2 + y^2 + z^2 = 36$ inside the cylinder $x^2 + y^2 = 6y$ and above the xy-plane.

▮ $x^2 + y^2 = 6y$ is equivalent to $x^2 + (y-3)^2 = 9$. So the cylinder has axis $x = 0$, $y = 3$, and radius 3. The base $\mathcal{R}$ is the circle $x^2 + (y-3)^2 = 9$. (See Fig. 44-32.) $S = \iint_{\mathcal{R}} \sqrt{1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2} \, dA$. $2x + 2z\dfrac{\partial z}{\partial x} = 0$. Hence, $\dfrac{\partial z}{\partial x} = -\dfrac{x}{z}$. Similarly, $\dfrac{\partial z}{\partial y} = -\dfrac{y}{z}$. Thus, $1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2 = 1 + \dfrac{x^2}{z^2} + \dfrac{y^2}{z^2} = \dfrac{x^2 + y^2 + z^2}{z^2} = \dfrac{36}{z^2}$. Therefore, $S = \iint_{\mathcal{R}} \dfrac{6}{z} \, dA$. Now use polar coordinates. The circle $x^2 + y^2 = 6y$ is equivalent to $r = 6\sin\theta$. $x^2 + y^2 + z^2 = 36$ is equivalent to $r^2 + z^2 = 36$, or $z^2 = 36 - r^2$. Hence, $S = \int_0^{\pi} \int_0^{6\sin\theta} \dfrac{6}{\sqrt{36 - r^2}} \, r \, dr \, d\theta = \int_0^{\pi} -6\sqrt{36 - r^2} \Big]_0^{6\sin\theta} d\theta = -6 \int_0^{\pi} (6|\cos\theta| - 6) \, d\theta = -12 \int_0^{\pi/2} (6\cos\theta - 6) \, d\theta$, since $|\cos(\pi - \theta)| = |\cos\theta|$. Hence, $S = -72(\sin\theta - \theta) \Big]_0^{\pi/2} = 72(\pi/2 - 1)$.

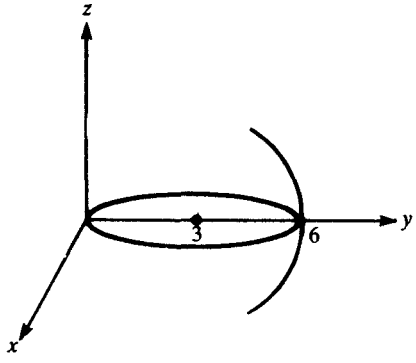


Fig. 44-32

44.55 Find the surface area S of the part of the sphere $x^2 + y^2 + z^2 = 4z$ inside the paraboloid $z = x^2 + y^2$.

▮ The region $\mathcal{R}$ under the spherical cap (Fig. 44-33) is obtained by finding the intersection of $x^2 + y^2 + z^2 = 4z$ and $z = x^2 + y^2$. This gives $z(z-3) = 0$. Hence, the paraboloid cuts the sphere when $z = 3$, and $\mathcal{R}$ is the disk $x^2 + y^2 \le 3$. $S = \iint_{\mathcal{R}} \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$. $2x + 2z\frac{\partial z}{\partial x} = 4\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial x} = -\frac{x}{z-2}$. Similarly, $\frac{\partial z}{\partial y} = -\frac{y}{z-2}$. Hence,

$$1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 1 + \frac{x^2}{(z-2)^2} + \frac{y^2}{(z-2)^2} = \frac{(z-2)^2 + x^2 + y^2}{(z-2)^2} = \frac{(x^2+y^2+z^2) - 4z + 4}{(z-2)^2} = \frac{4}{(z-2)^2}$$

Therefore,

$$S = \iint_{\mathcal{R}} \frac{2}{z-2}\, dA = \int_0^{2\pi}\int_0^{\sqrt{3}} \frac{2}{\sqrt{4-r^2}}\, r\, dr\, d\theta = -\int_0^{2\pi} 2\sqrt{4-r^2}\,\Big]_0^{\sqrt{3}} d\theta = -2\int_0^{2\pi}(1-2)\, d\theta = 2\cdot 2\pi = 4\pi$$

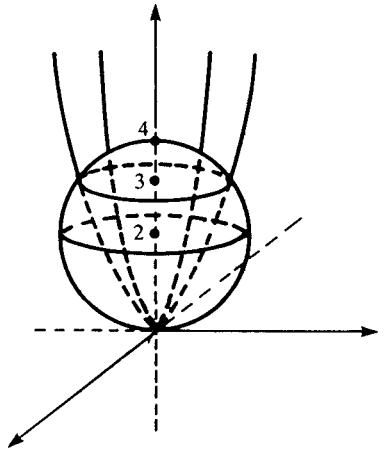


Fig. 44-33

44.56 Find the surface area of the part of the sphere $x^2 + y^2 + z^2 = 25$ between the planes $z = 2$ and $z = 4$.

▮ The surface lies above the ring-shaped region $\mathcal{R}$: $3 \le r \le \sqrt{21}$. $S = \iint_{\mathcal{R}} \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA = \iint_{\mathcal{R}} \sqrt{1 + \frac{x^2}{z^2} + \frac{y^2}{z^2}}\, dA = \iint_{\mathcal{R}} \sqrt{\frac{25}{z^2}}\, dA = \iint_{\mathcal{R}} \frac{5}{z}\, dA = \int_0^{2\pi}\int_3^{\sqrt{21}} \frac{5}{\sqrt{25-r^2}}\, r\, dr\, d\theta$, since $r^2 + z^2 = 25$. Hence, $S = \int_0^{2\pi} -5\sqrt{25-r^2}\,\big]_3^{\sqrt{21}}\, d\theta = -5\int_0^{2\pi}(2-4)\, d\theta = 10\int_0^{2\pi} d\theta = 20\pi$.

44.57 Find the surface area of a sphere of radius a.

▮ Consider the upper hemisphere of the sphere $x^2 + y^2 + z^2 = a^2$. It projects down onto the disk $\mathcal{R}$ of radius a whose center is at the origin. Hence, the surface area of the entire sphere is $2\iint_{\mathcal{R}} \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA = 2\iint_{\mathcal{R}} \sqrt{1 + \frac{x^2}{z^2} + \frac{y^2}{z^2}}\, dA = 2\iint_{\mathcal{R}} \sqrt{\frac{x^2+y^2+z^2}{z^2}}\, dA = 2\iint_{\mathcal{R}} \frac{a}{z}\, dA = 2\int_0^{2\pi}\int_0^{a} \frac{a}{\sqrt{a^2-r^2}}\, r\, dr\, d\theta =$ $2a\int_0^{2\pi}(-\sqrt{a^2-r^2})\,\big]_0^a\, d\theta = -2a\int_0^{2\pi}(-a)\, d\theta = 2a^2\int_0^{2\pi} d\theta = 2a^2\cdot 2\pi = 4\pi a^2$.

44.58 Find the surface area of a cone of height h and radius of base b.

▮ Consider the cone $z = \frac{h}{b}\sqrt{x^2+y^2}$, or $b^2z^2 = h^2x^2 + h^2y^2$ (see Fig. 44-30). The portion of the cone under $z = h$ projects onto the interior $\mathcal{R}$ of the circle $r = b$ in the xy-plane. Then

$$S = \iint_{\mathcal{R}} \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA = \iint_{\mathcal{R}} \sqrt{1 + \left(\frac{h^2 x}{b^2 z}\right)^2 + \left(\frac{h^2 y}{b^2 z}\right)^2}\, dA = \iint_{\mathcal{R}} \sqrt{\frac{b^4 z^2 + h^4 x^2 + h^4 y^2}{b^4 z^2}}\, dA$$

$$= \iint_{\mathcal{R}} \sqrt{\frac{h^2(b^2 + h^2)(x^2 + y^2)}{b^4 z^2}}\, dA = \iint_{\mathcal{R}} \frac{h}{b^2}\sqrt{b^2 + h^2}\,\frac{\sqrt{x^2 + y^2}}{z}\, dA = \iint_{\mathcal{R}} \frac{h}{b^2}\sqrt{b^2 + h^2}\cdot\frac{b}{h}\, dA$$

$$= \frac{1}{b}\sqrt{b^2 + h^2}\iint_{\mathcal{R}} dA = \frac{\sqrt{b^2 + h^2}}{b}(\pi b^2) = \pi b\sqrt{b^2 + h^2} = \pi b s$$

where $s = \sqrt{b^2 + h^2}$ is the slant height of the cone.

44.59 Use a triple integral to find the volume V inside $x^2 + y^2 = 9$, above $z = 0$, and below $x + z = 4$.

▮ $V = \int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}\int_{0}^{4-x} 1\, dz\, dy\, dx$. The bounds on z correspond to the requirement that the solid is above $z = 0$ and below $x + z = 4$. The bounds on y come from the equation $x^2 + y^2 = 9$. $V = \int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} z\,\Big]_0^{4-x} dy\, dx = \int_{-3}^{3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}(4-x)\, dy\, dx = \int_{-3}^{3}(4-x)y\,\Big]_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx = \int_{-3}^{\sqrt{3}} 2(4-x)\sqrt{9-x^2}\, dx = 8\int_{-3}^{3}\sqrt{9-x^2}\, dx + \int_{-3}^{3}(-2x)\sqrt{9-x^2}\, dx$. Now, $\int_{-3}^{3}\sqrt{9-x^2}\, dx$ gives the area of the upper half of the disk $x^2 + y^2 \le 9$ of radius 3, namely, $\frac{1}{2}(9\pi) = \frac{9}{2}\pi$. Also, $\int_{-3}^{3}(-2x)\sqrt{9-x^2}\, dx = 0$ (odd function). Therefore, $V = 8(\frac{9}{2}\pi) = 36\pi$.

44.60 Use a triple integral to find the volume V inside $x^2 + y^2 = 4x$, above $z = 0$, and below $x^2 + y^2 = 4z$.

▮ $x^2 + y^2 = 4x$ is equivalent to $(x-2)^2 + y^2 = 4$, the cylinder of radius 2 with axis $x = 2$, $y = 0$. Hence, $V = \int_0^4\int_{-\sqrt{4x-x^2}}^{\sqrt{4x-x^2}}\int_0^{(x^2+y^2)/4} 1\, dz\, dy\, dx$. This is difficult to compute; so let us switch to cylindrical coordinates. The circle $x^2 + y^2 = 4x$ becomes $r = 4\cos\theta$, and we get $V = \int_0^{\pi}\int_0^{4\cos\theta}\int_0^{r^2/4} r\, dz\, dr\, d\theta = \int_0^{\pi}\int_0^{4\cos\theta} rz\,\Big]_0^{r^2/4} dr\, d\theta = \int_0^{\pi}\int_0^{4\cos\theta}\frac{1}{4}r^3\, dr\, d\theta = \int_0^{\pi}\frac{1}{16}r^4\,\Big]_0^{4\cos\theta} d\theta = \int_0^{\pi} 16\cos^4\theta\, d\theta = 32\int_0^{\pi/2}\cos^4\theta\, d\theta = 6\pi$ (Problem 44.29).

44.61 Use a triple integral to find the volume inside the cylinder $r = 4$, above $z = 0$, and below $2z = y$.

▮ The solid is wedge-shaped. The base is the half-disk $0 \le \theta \le \pi$, $0 \le r \le 4$. The height is $\frac{y}{2} = \frac{r\sin\theta}{2}$. Then

$$V = \int_0^{\pi}\int_0^4\int_0^{(r\sin\theta)/2} r\, dz\, dr\, d\theta = \int_0^{\pi}\int_0^4 rz\,\Big]_0^{(r\sin\theta)/2} dr\, d\theta = \int_0^{\pi}\int_0^4\frac{1}{2}r^2\sin r\, dr\, d\theta = \frac{1}{6}\int_0^{\pi} r^3\sin\theta\,\Big]_0^4 d\theta$$

$$= \frac{1}{6}\int_0^{\pi} 64\sin\theta\, d\theta = \frac{32}{3}(-\cos\theta)\,\Big]_0^{\pi} = -\frac{32}{3}(-1-1) = \frac{64}{3}$$

44.62 Use a triple integral to find the volume cut from the cone $\phi = \pi/4$ by the sphere $\rho = 2a\cos\phi$.

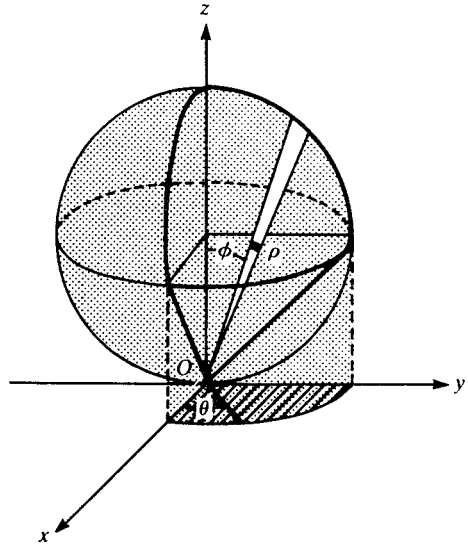


Fig. 44-34

▮ Refer to Fig. 44-34. Note that $\rho = 2a\cos\phi$ has the equivalent forms $\rho^2 = 2a\rho\cos\phi$, $x^2 + y^2 + z^2 = 2az$, $x^2 + y^2 + (z-a)^2 = a^2$. Thus, it is the sphere with center at $(0, 0, a)$ and radius a. Then, $V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{2a\cos\phi} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi}\int_0^{\pi/4} \frac{1}{3}\rho^3 \sin\phi \Big]_0^{2a\cos\phi} d\phi \, d\theta = \int_0^{2\pi}\int_0^{\pi/4} \frac{8a^3}{3} \sin\phi\cos^3\phi \, d\phi \, d\theta = \frac{8a^3}{3}\int_0^{2\pi}\left(-\frac{\cos^4\phi}{4}\right)\Big]_0^{\pi/4} d\theta = -\frac{2a^3}{3}\int_0^{2\pi}\left(\frac{1}{4} - 1\right) d\theta = \frac{1}{2}a^3\int_0^{2\pi} d\theta = \frac{1}{2}a^3 \cdot 2\pi = \pi a^3$.

44.63 Find the volume within the cylinder $r = 4\cos\theta$ bounded above by the sphere $r^2 + z^2 = 16$ and below by the plane $z = 0$.

▮ Integrate first with respect to z from $z = 0$ to $z = \sqrt{16 - r^2}$, then with respect to r from $r = 0$ to $r = 4\cos\theta$, and then with respect to θ from $\theta = 0$ to $\theta = \pi$. (See Fig. 44-35.) $V = \int_0^{\pi}\int_0^{4\cos\theta}\int_0^{\sqrt{16-r^2}} r \, dz \, dr \, d\theta = \int_0^{\pi}\int_0^{4\cos\theta} rz \Big]_0^{\sqrt{16-r^2}} dr \, d\theta = \int_0^{\pi}\int_0^{4\cos\theta} r\sqrt{16 - r^2} \, dr \, d\theta = \int_0^{\pi}[-\frac{1}{2}\cdot\frac{2}{2}(16 - r^2)^{3/2}]\Big]_0^{4\cos\theta} d\theta = -\frac{1}{3}\int_0^{\pi}(64\sin^3\theta - 64)\, d\theta = -\frac{64}{3}\int_0^{\pi}(\sin\theta - \cos^2\theta\sin\theta - 1)\, d\theta = -\frac{64}{3}(-\cos\theta + \frac{1}{3}\cos^3\theta - \theta)\Big]_0^{\pi} = -\frac{64}{3}[(1 - \frac{1}{3} - \pi) - (-1 + \frac{1}{3})] = -\frac{64}{3}(\frac{4}{3} - \pi) = \frac{64}{9}(3\pi - 4)$.

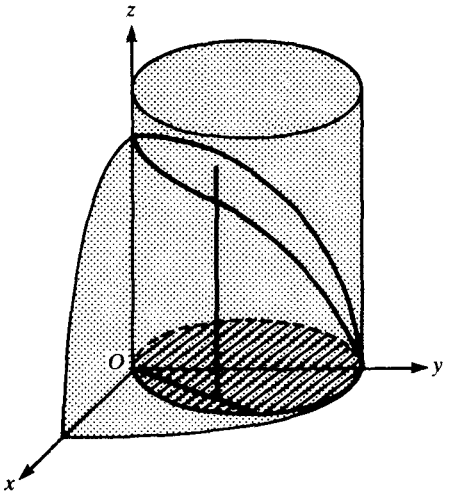


Fig. 44-35

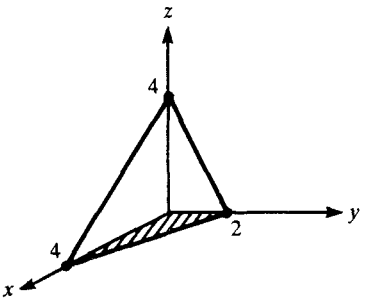


Fig. 44-36

44.64 Evaluate $I = \iiint_{\mathcal{R}} x \, dV$, where $\mathcal{R}$ is the tetrahedron bounded by the coordinate planes and the plane $x + 2y + z = 4$ (see Fig. 44-36).

▮ y can be integrated from 0 to 2. In the base triangle, for a given y, x runs from 0 to $4 - 2y$. For given x and y, z varies from 0 to $4 - x - 2y$. Hence, $I = \int_0^2\int_0^{4-2y}\int_0^{4-x-2y} x \, dz \, dx \, dy = \int_0^2\int_0^{4-2y} xz \Big]_0^{4-x-2y} dx \, dy = \int_0^2\int_0^{4-2y} x(4 - x - 2y)\, dx \, dy = \int_0^2\int_0^{4-2y}(4x - x^2 - 2xy)\, dx \, dy = \int_0^2(2x^2 - \frac{1}{3}x^3 - yx^2)\Big]_0^{4-2y} dy = \int_0^2(4 - 2y)^2[2 - \frac{1}{3}(4 - 2y) - y]\, dy = \int_0^2(4 - 2y)^2(\frac{2}{3} - \frac{1}{3}y)\, dy = \frac{4}{3}\int_0^2(2 - y)^3\, dy = \frac{4}{3}\cdot(-\frac{1}{4})(2 - y)^4\Big]_0^2 = -\frac{1}{3}(-16) = \frac{16}{3}$.

44.65 Evaluate $I = \iiint_{\mathcal{R}} (x^2 + y^2 + z^2)\, dV$, where $\mathcal{R}$ is the ball of radius a with center at the origin.

▮ Use spherical coordinates. $I = \int_0^{2\pi}\int_0^{\pi}\int_0^a \rho^2\cdot\rho^2\sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi}\int_0^{\pi}\frac{1}{5}\rho^5\sin\phi\Big]_0^a d\phi \, d\theta = \frac{1}{5}a^5\int_0^{2\pi}\int_0^{\pi}\sin\phi \, d\phi \, d\theta = \frac{1}{5}a^5\int_0^{2\pi}(-\cos\phi)\Big]_0^{\pi} d\theta = \frac{1}{5}a^5\int_0^{2\pi}[1 - (-1)]\, d\theta = \frac{2}{5}a^5\int_0^{2\pi} d\theta = \frac{2}{5}a^5\cdot 2\pi = \frac{4}{5}\pi a^5$.

44.66 Use a triple integral to find the volume V of the solid inside the cylinder $x^2 + y^2 = 25$ and between the planes $z = 2$ and $x + z = 8$.

▮ The projection on the xy-plane is the circle $x^2 + y^2 = 25$. Use cylindrical coordinates.

$$V = \int_0^{2\pi}\int_0^5\int_2^{8-(r\cos\theta)} r \, dz \, dr \, d\theta = \int_0^{2\pi}\int_0^5 rz\Big]_2^{8-(r\cos\theta)} dr \, d\theta = \int_0^{2\pi}\int_0^5 r(6 - r\cos\theta)\, dr \, d\theta$$

$$= \int_0^{2\pi}(3r^2 - \tfrac{1}{3}r^3\cos\theta)\Big]_0^5 d\theta = \int_0^{2\pi} 25(3 - \tfrac{5}{3}\cos\theta)\, d\theta = 25(3\theta - \tfrac{5}{3}\sin\theta)\Big]_0^{2\pi} = 25(6\pi) = 150\pi.$$

44.67 Find the volume of the solid enclosed by the paraboloids $z = x^2 + y^2$ (upward-opening) and $z = 36 - 3x^2 - 8y^2$ (downward-opening).

I The projection of the intersection of the surfaces is the ellipse $4x^2 + 9y^2 = 36$. By symmetry, we can calculate the integral with respect to x and y in the first quadrant and then multiply it by 4.

$$V = 4\int_0^3 \int_0^{2\sqrt{9-x^2}/3} \int_{x^2+y^2}^{36-3x^2-8y^2} dz\,dy\,dx = 4\int_0^3 \int_0^{2\sqrt{9-x^2}/3} [(36-3x^2-8y^2)-(x^2+y^2)]\,dy\,dx$$

$$= 4\int_0^3 \int_0^{2\sqrt{9-x^2}/3} (36-4x^2-9y^2)\,dy\,dx = 4\int_0^3 (36y - 4x^2y - 3y^3)\Big]_0^{2\sqrt{9-x^2}/3} dx$$

$$= \frac{8}{3}\int_0^3 \sqrt{9-x^2}\,[36-4x^2-4(9-x^2)]\,dx = \frac{32}{3}\int_0^3 \sqrt{9-x^2}\left[\frac{2}{3}(9-x^2)\right] dx = \frac{64}{9}\int_0^3 (9-x^2)^{3/2}\,dx$$

Let $x = 3\sin\theta$, $dx = 3\cos\theta\,d\theta$. Then $V = \frac{64}{9}\int_0^{\pi/2} 27\cos^3\theta \cdot 3\cos\theta\,d\theta = 576\int_0^{\pi/2} \cos^4\theta\,d\theta = 108\pi$ (from Problem 44.29).

44.68 Describe the solid whose volume is given by the integral $\int_0^2 \int_{2x}^4 \int_0^1 dz\,dy\,dx$.

I See Fig. 44-37. The solid lies under the plane $z = 1$ and above the region in the first quadrant of the xy-plane bounded by the lines $y = 2x$ and $y = 4$.

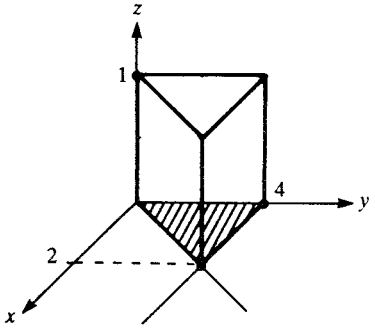


Fig. 44-37

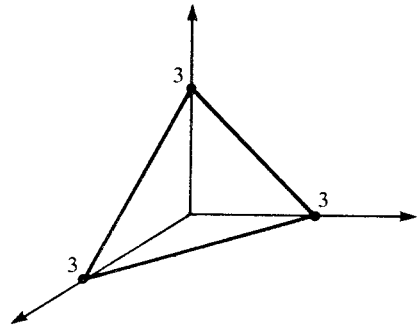


Fig. 44-38

44.69 Describe the solid whose volume is given by the integral $\int_0^3 \int_0^{3-x} \int_0^{3-x-y} dz\,dy\,dx$.

I See Fig. 44-38. The solid lies under the plane $z = 3 - x - y$ and above the triangle in the xy-plane bounded by the coordinate axes and the line $x + y = 3$. It is a tetrahedron, of volume $\frac{1}{6}(3)^3 = \frac{9}{2}$.

44.70 Describe the solid whose volume is given by the integral $\int_0^5 \int_0^{\sqrt{25-x^2}} \int_0^3 dz\,dy\,dx$, and compute the volume.

I The solid is the part of the solid right circular cylinder $x^2 + y^2 \le 25$ lying in the first octant between $z = 0$ and $z = 3$ (see Fig. 44-39). Its volume is $\frac{1}{4}(\pi r^2 h) = \frac{1}{4}[\pi(5)^2 \cdot 3] = \frac{75}{4}\pi$.

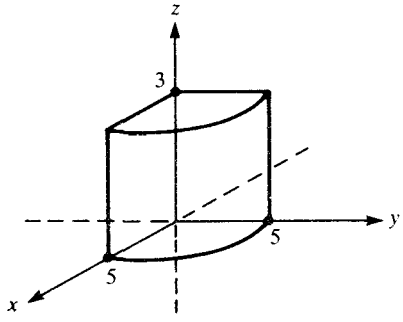


Fig. 44-39

44.71 If $\mathscr{R}$ is a rectangular box $x_1 \le x \le x_2$, $y_1 \le y \le y_2$, $z_1 \le z \le z_2$, show that $I = \iiint_{\mathscr{R}} f(x)g(u)h(z)\,dV = \underbrace{(\int_{x_1}^{x_2} f(x)\,dx)}_{a}\underbrace{(\int_{y_1}^{y_2} g(y)\,dy)}_{b}\underbrace{(\int_{z_1}^{z_2} h(z)\,dz)}_{c}$.

I $I = \int_{z_1}^{z_2}\int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x)g(y)h(z)\,dx\,dy\,dz = \int_{z_1}^{z_2}\int_{y_1}^{y_2} g(y)\,h(z)[\int_{x_1}^{x_2} f(x)\,dx]\,dy\,dz = a\int_{z_1}^{z_2} h(z)[\int_{y_1}^{y_2} g(y)\,dy]\,dz = ab\int_{z_1}^{z_2} h(z)\,dz = abc$.

44.72 Evaluate $I = \iiint_{\mathscr{R}} x^3 y e^z\,dV$, where $\mathscr{R}$ is the rectangular box: $1 \le x \le 2$, $0 \le y \le 1$, $0 \le z \le \ln 2$.

I By Problem 44.71, $I = \int_1^2 x^3\,dx \cdot \int_0^1 y\,dy \cdot \int_0^{\ln 2} e^z\,dz = \frac{1}{4}x^4\Big]_1^2 \cdot \frac{1}{2}y^2\Big]_0^1 \cdot e^z\Big]_0^{\ln 2} = (4 - \frac{1}{4}) \cdot \frac{1}{2} \cdot (2-1) = \frac{15}{8}$.

44.73 Evaluate $J = \iiint_{\mathscr{R}} x^2\,dV$ for the ball $\mathscr{R}$ of Problem 44.65.

▮ By spherical symmetry, $I \equiv \iiint_{\mathscr{R}} x^2\,dV + \iiint_{\mathscr{R}} y^2\,dV + \iiint_{\mathscr{R}} z^2\,dV = J + J + J$; so $J = \frac{1}{3}I = \dfrac{4\pi a^5}{15}$.

44.74 Find the mass of a plate in the form of a right triangle $\mathscr{R}$ with legs a and b, if the density (mass per unit area) is numerically equal to the sum of the distances from the legs.

▮ Let the right angle be at the origin and the legs a and b be along the positive x and y axes, respectively (Fig. 44-40). The density $\delta(x, y) = x + y$. Hence, the mass $M = \iint (x + y)\,dA = \int_0^a \int_0^{b-(b/a)x} (x + y)\,dy\,dx = \int_0^a (xy + \frac{1}{2}y^2)\,]_0^{b-(b/a)x}\,dx = \dfrac{b}{a}\int_0^a (a - x)[x + \frac{1}{2}(b/a)(a - x)]\,dx = \dfrac{b}{a}\int_0^a [ax - x^2 + \frac{1}{2}(b/a)(a - x)^2]\,dx = (b/a)[\frac{1}{2}ax^2 - \frac{1}{3}x^3 - \frac{1}{2}(b/a)\cdot\frac{1}{3}(a - x)^3]\,]_0^a = (b/a)\{(\frac{1}{2}a^3 - \frac{1}{3}a^3) - [-\frac{1}{2}(b/a)\cdot\frac{1}{3}a^3]\} = (b/a)(\frac{1}{6}a^3 + \frac{1}{6}ba^2) = \frac{1}{6}ba(a + b)$.

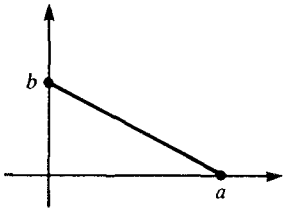


Fig. 44-40

44.75 Find the mass of a circular plate $\mathscr{R}$ of radius a whose density is numerically equal to the distance from the center.

▮ Let the circle be $r = a$. Then $M = \iint_{\mathscr{R}} r\,dA = \int_0^{2\pi}\int_0^a r\cdot r\,dr\,d\theta = \int_0^{2\pi} \frac{1}{3}r^3\,]_0^a\,d\theta = \int_0^{2\pi} \frac{1}{3}a^3\,d\theta = \frac{1}{3}a^3\cdot 2\pi = \frac{2}{3}\pi a^3$.

44.76 Find the mass of a solid right circular cylinder $\mathscr{R}$ of height h and radius of base b, if the density (mass per unit volume) is numerically equal to the square of the distance from the axis of the cylinder.

▮ $M = \iiint_{\mathscr{R}} r^2\,dV = \int_0^{2\pi}\int_0^b\int_0^h r^2\cdot r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^b r^3h\,dr\,d\theta = \int_0^{2\pi} \frac{1}{4}hr^4\,]_0^b\,d\theta = \int_0^{2\pi} \frac{1}{4}hb^4\,d\theta = \frac{1}{4}hb^4\cdot 2\pi = \frac{1}{2}\pi hb^4$.

44.77 Find the mass of a ball $\mathscr{B}$ of radius a whose density is numerically equal to the distance from a fixed diametral plane.

▮ Let the ball be the inside of the sphere $x^2 + y^2 + z^2 = a^2$, and let the fixed diametral plane be $z = 0$. Then $M = \iiint |z|\,dV$. Use the upper hemisphere and double the result. In spherical coordinates,

$$M = 2\int_0^{2\pi}\int_0^{2\pi}\int_0^a z\cdot\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = 2\int_0^{2\pi}\int_0^{\pi/2}\int_0^a \rho\cos\phi\cdot\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$$
$$= 2\int_0^{2\pi}\int_0^{\pi/2} \tfrac{1}{4}\rho^4\cos\phi\sin\phi\,]_0^a\,d\phi\,d\theta = \tfrac{1}{2}a^4\int_0^{2\pi}\tfrac{1}{2}\sin^2\phi\,]_0^{\pi/2}\,d\theta = \tfrac{1}{4}a^4\int_0^{2\pi}d\theta = \tfrac{1}{4}a^4\cdot 2\pi = \tfrac{1}{2}\pi a^4$$

44.78 Find the mass of a solid right circular cone $\mathscr{C}$ of height h and radius of base b whose density is numerically equal to the distance from its axis.

▮ Let $\alpha = \tan^{-1}(b/h)$. In spherical coordinates, the lateral surface is $\phi = \alpha$. $M = \iiint_{\mathscr{C}} r\,dV = \int_0^{2\pi}\int_0^{\alpha}\int_0^{h\sec\phi} \rho\sin\phi\cdot\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \int_0^{2\pi}\int_0^{\alpha} \tfrac{1}{4}\rho^4\sin^2\phi\,]_0^{h\sec\phi}\,d\phi\,d\theta = \tfrac{1}{4}h^4\int_0^{2\pi}\int_0^{\alpha}\sec^4\phi\sin^2\phi\,d\phi\,d\theta = \tfrac{1}{4}h^4\int_0^{2\pi}\int_0^{\alpha}\sec^2\phi\tan^2\phi\,d\phi\,d\theta = \tfrac{1}{4}h^4\int_0^{2\pi}\tfrac{1}{3}\tan^3\phi\,]_0^{\alpha}\,d\theta = \tfrac{1}{4}h^4\cdot\tfrac{1}{3}\tan^3\alpha\int_0^{2\pi}d\theta = \tfrac{1}{12}h^4(b/h)^3\cdot 2\pi = \tfrac{1}{6}\pi hb^3$.

44.79 Find the mass of a spherical surface $\mathscr{S}$ whose density is equal to the distance from a fixed diametral plane.

▮ Let $\mathscr{S}$ be $x^2 + y^2 + z^2 = a^2$, and let the fixed diametral plane be the xy-plane. Then $M = \iint_{\mathscr{C}} |z|\,\dfrac{dS}{dA}\,dA$, where $\mathscr{C}$ is the disk $x^2 + y^2 = a^2$. Then, if we double the mass of the upper hemisphere, $M = 2\iint_{\mathscr{C}} z\sqrt{1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2}\,dA$. $\dfrac{\partial z}{\partial x} = -\dfrac{x}{z}$ and $\dfrac{\partial z}{\partial y} = -\dfrac{y}{z}$. Then $1 + \left(\dfrac{\partial z}{\partial x}\right)^2 + \left(\dfrac{\partial z}{\partial y}\right)^2 = 1 + \dfrac{x^2}{z^2} + \dfrac{y^2}{z^2} = \dfrac{x^2 + y^2 + z^2}{z^2} = \dfrac{a^2}{z^2}$. Hence, $M = 2\iint_{\mathscr{C}} a\,dA = 2a\iint_{\mathscr{C}} dA = 2a(\text{area of a circle of radius } a) = 2a(\pi a^2) = 2\pi a^3$.

44.80 Find the center of mass $(\bar{x}, \bar{y})$ of the plate cut from the parabola $y^2 = 8x$ by its latus rectum $x = 2$ if the density is numerically equal to the distance from the latus rectum.

▮ See Fig. 44-41. By symmetry, $\bar{y} = 0$. The mass $M = \iint (2 - x)\, dA = \int_{-4}^{4} \int_{y^2/8}^{2} (2 - x)\, dx\, dy = \int_{-4}^{4} \left(2x - \frac{1}{2}x^2\right)\Big]_{y^2/8}^{2} dy = \int_{-4}^{4} \left[2 - \left(\frac{y^2}{4} - \frac{y^4}{128}\right)\right] dy = \left(2y - \frac{1}{12}y^3 + \frac{1}{128}\frac{y^5}{5}\right)\Big]_{-4}^{4} = 8\left(2 - \frac{16}{12} + \frac{1}{128}\cdot\frac{256}{5}\right) = \frac{128}{15}$. The moment about the y-axis is given by $M_y = \int_{-4}^{4} \int_{y^2/8}^{2} x(2 - x)\, dx\, dy = \int_{-4}^{4} \left(x^2 - \frac{1}{3}x^3\right)\Big]_{y^2/8}^{2} dy = \int_{-4}^{4} \left[\frac{4}{3} - \left(\frac{y^4}{64} - \frac{y^6}{3\cdot 512}\right)\right] dy = \left(\frac{4y}{3} - \frac{1}{64}\frac{y^5}{5} + \frac{1}{3\cdot 512}\frac{y^7}{7}\right)\Big]_{-4}^{4} = 8\left(\frac{4}{3} - \frac{4}{5} + \frac{8}{21}\right) = \frac{256}{35}$. Hence, $\bar{x} = \frac{M_y}{M} = \frac{\frac{256}{35}}{\frac{128}{15}} = \frac{6}{7}$. Thus, the center of mass is $\left(\frac{6}{7}, 0\right)$.

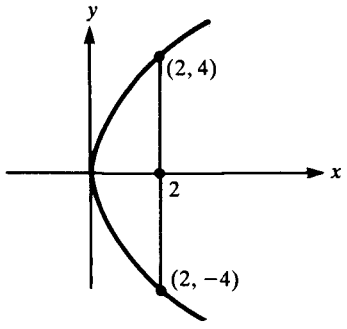


Fig. 44-41

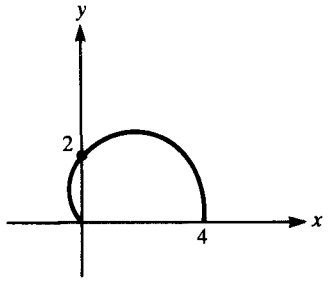


Fig. 44-42

44.81 Find the center of mass of a plate in the form of the upper half of the cardioid $r = 2(1 + \cos\theta)$ if the density is numerically equal to the distance from the pole.

▮ See Fig. 44-42. The mass $M = \iint_{\mathcal{R}} r\, dA = \int_0^{\pi} \int_0^{2(1+\cos\theta)} r \cdot r\, dr\, d\theta = \int_0^{\pi} \frac{1}{3}r^3\Big]_0^{2(1+\cos\theta)} d\theta = \frac{8}{3}\int_0^{\pi} (1 + \cos\theta)^3\, d\theta = \frac{8}{3}\int_0^{\pi} \left[1 + 3\cos\theta + 3\left(\frac{1 + \cos 2\theta}{2}\right) + (\cos\theta - \sin^2\theta\cos\theta)\right] d\theta = \frac{8}{3}\left(\frac{5}{2}\theta + 4\sin\theta + \frac{3}{4}\sin 2\theta - \frac{1}{3}\sin^3\theta\right)\Big]_0^{\pi} = \frac{8}{3}\left(\frac{5}{2}\cdot\pi\right) = \frac{20\pi}{3}$. The moment about the x-axis is $M_x = \int_0^{\pi} \int_0^{2(1+\cos\theta)} yr^2\, dr\, d\theta = \int_0^{\pi} \int_0^{2(1+\cos\theta)} r^3 \sin\theta\, dr\, d\theta = \int_0^{\pi} \frac{1}{4}r^4 \sin\theta\Big]_0^{2(1+\cos\theta)} d\theta = 4\int_0^{\pi} (1 + \cos\theta)^4 \sin\theta\, d\theta = -4\frac{(1 + \cos\theta)^5}{5}\Big]_0^{\pi} = -\frac{4}{5}(-32) = \frac{128}{5}$. Hence, $\bar{y} = \frac{M_x}{M} = \frac{128/5}{20\pi/3} = \frac{96}{25\pi}$. The moment about the y-axis is $M_y = \int_0^{\pi} \int_0^{2(1+\cos\theta)} xr^2\, dr\, d\theta = \int_0^{\pi} \int_0^{2(1+\cos\theta)} r^3 \cos\theta\, dr\, d\theta = \int_0^{\pi} \frac{1}{4}r^4 \cos\theta\Big]_0^{2(1+\cos\theta)} d\theta = 4\int_0^{\pi} (1 + \cos\theta)^4 \cos\theta\, d\theta = 4\int_0^{\pi} \left[\cos\theta + 2(1 + \cos 2\theta) + 6(\cos\theta - \sin^2\theta\cos\theta) + \left(1 + 2\cos 2\theta + \frac{1 + \cos 4\theta}{2}\right) + (\cos\theta - 2\sin^2\theta\cos\theta + \sin^4\theta\cos\theta)\right] d\theta = 4(\frac{7}{2}\theta + 8\sin\theta + 2\sin 2\theta + \frac{1}{8}\sin 4\theta - \frac{8}{3}\sin^3\theta + \frac{1}{5}\sin^5\theta)\Big]_0^{\pi} = 4(\frac{7}{2}\pi) = 14\pi$. Hence, $\bar{x} = \frac{M_y}{M} = \frac{14\pi}{20\pi/3} = \frac{21}{10}$. So the center of mass is $\left(\frac{21}{10}, \frac{96}{25\pi}\right)$.

44.82 Find the center of mass of the first-quadrant part of the disk of radius a with center at the origin, if the density function is y.

▮ The mass $M = \int_0^a \int_0^{\sqrt{a^2-y^2}} y\, dx\, dy = \int_0^a y\sqrt{a^2 - y^2}\, dy = -\frac{1}{2}\cdot\frac{2}{3}(a^2 - y^2)^{3/2}\Big]_0^a = -\frac{1}{3}(-a^3) = \frac{1}{3}a^3$. The moment about the x-axis is

$$M_x = \iint_{\mathcal{R}} y \cdot y\, dA = \int_0^{\pi/2} \int_0^a y^2 \cdot r\, dr\, d\theta = \int_0^{\pi/2} \int_0^a r^3 \sin\theta\, dr\, d\theta = \int_0^{\pi/2} \frac{1}{4}a^4 \sin^2\theta\, d\theta$$

$$= \frac{1}{4}a^4 \int_0^{\pi/2} \frac{1 - \cos 2\theta}{2}\, d\theta = \frac{1}{8}a^4\left(\theta - \frac{1}{2}\sin 2\theta\right)\Big]_0^{\pi/2} = \frac{a^4}{8}\left(\frac{\pi}{2}\right) = \frac{\pi a^4}{16}$$

Hence, $\bar{y} = \frac{M_x}{M} = \frac{\pi a^4/16}{a^3/3} = \frac{3\pi a}{16}$. The moment about the y-axis is $M_y = \int_0^a \int_0^{\sqrt{a^2-y^2}} xy\, dx\, dy = \int_0^a \frac{1}{2}yx^2\Big]_0^{\sqrt{a^2-y^2}} dy = \frac{1}{2}\int_0^a y(a^2 - y^2)\, dy = \frac{1}{2}\left(\frac{a^2}{2}y^2 - \frac{y^4}{4}\right)\Big]_0^a = \frac{1}{2}\left(\frac{a^4}{2} - \frac{a^4}{4}\right) = \frac{a^4}{8}$. Therefore, $\bar{x} = \frac{M_y}{M} = \frac{a^4/8}{a^3/3} = \frac{3a}{8}$. Hence, the center of mass is $\left(\frac{3a}{8}, \frac{3\pi a}{16}\right)$.

44.83 Find the center of mass of the cube of edge a with three faces on the coordinate planes, if the density is numerically equal to the sum of the distances from the coordinate planes.

▮ The mass $M = \int_0^a \int_0^a \int_0^a (x+y+z)\,dx\,dy\,dz = 3(\int_0^a x\,dx)(\int_0^a dy)(\int_0^a dz) = \frac{3}{2}a^4$. The moment about the xz-plane is $M_{xz} = \int_0^a \int_0^a \int_0^a y(x+y+z)\,dx\,dy\,dz = \int_0^a \int_0^a y\left[\frac{1}{2}x^2 + (y+z)x\right]\Big]_0^a dy\,dz = \int_0^a \int_0^a y\left[\frac{a^2}{2} + a(y+z)\right] dy\,dz = \int_0^a \left[\frac{a^2}{4}y^2 + a\left(\frac{1}{3}y^3 + \frac{1}{2}zy^2\right)\right]\Big]_0^a dz = \int_0^a \left(\frac{a^4}{4} + \frac{a^4}{3} + \frac{a^3}{2}z\right) dz = \frac{7}{12}a^4 z + \frac{a^3}{4}z^2\Big]_0^a = \frac{7}{12}a^5 + \frac{1}{4}a^5 = \frac{5}{6}a^5$. So $\bar{y} = \frac{M_{xz}}{M} = \frac{\frac{5}{6}a^5}{\frac{3}{2}a^4} = \frac{5}{9}a$. By symmetry, $\bar{x} = \frac{5}{9}a$, $\bar{z} = \frac{5}{9}a$.

44.84 Find the center of mass of the first octant of the ball of radius a, $x^2 + y^2 + z^2 \le a^2$, if the density is numerically equal to z.

▮ The mass is one-eighth of that of the ball in Problem 44.77, or $\pi a^4/16$. The moment about the xz-plane is

$$M_{xz} = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^a \underbrace{\rho\sin\phi\sin\theta}_{y}\cdot\underbrace{\rho\cos\phi}_{z}\cdot\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \int_0^{\pi/2}\int_0^{\pi/2} \rho^5\sin^2\phi\cos\phi\sin\theta\Big]_0^a d\phi\,d\theta$$

$$= \frac{a^5}{5}\int_0^{\pi/2}\frac{1}{3}\sin^3\phi\sin\theta\Big]_0^{\pi/2} d\theta = \frac{a^5}{5}\int_0^{\pi/2}\frac{1}{3}\sin\theta\,d\theta = \frac{a^5}{15}(-\cos\theta)\Big]_0^{\pi/2} = \frac{a^5}{15}.$$

Hence, $\bar{y} = \frac{M_{xz}}{M} = \frac{a^5/15}{\pi a^4/16} = \frac{16a}{15\pi}$. By symmetry, $\bar{x} = \frac{16a}{15\pi}$. The moment about the xy-plane is $M_{xy} = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^a \rho^2\cos^2\theta\cdot\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \int_0^{\pi/2}\int_0^{\pi/2}\frac{1}{5}a^5\cos^2\phi\sin\theta\,d\phi\,d\theta = \frac{a^5}{5}\int_0^{\pi/2}\left(-\frac{\cos^3\phi}{3}\right)\Big]_0^{\pi/2} d\theta = \frac{a^5}{5}\int_0^{\pi/2}\frac{1}{3}\,d\theta = \frac{a^5}{15}\cdot\frac{\pi}{2} = \frac{\pi a^5}{30}$. Hence $\bar{z} = \frac{M_{xy}}{M} = \frac{\pi a^5/30}{\pi a^4/16} = \frac{8a}{15}$. Thus, the center of mass is $\left(\frac{16a}{15\pi}, \frac{16a}{15\pi}, \frac{8a}{15}\right)$.

44.85 Find the center of mass of a solid right circular cone $\mathscr{C}$ of height h and radius of base b, if the density is equal to the distance from the base.

▮ Let the cone have the base $x^2 + y^2 \le b$ in the xy-plane and vertex at $(0, 0, h)$; its equation is then $z = h - (h/b)r$. By symmetry, $\bar{x} = \bar{y} = 0$. The mass $M = \iiint_{\mathscr{C}} z\,dV$. Use cylindrical coordinates. Then $M = \int_0^{2\pi}\int_0^b \int_0^{h-(h/b)r} zr\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^b \frac{1}{2}rz^2\Big]_0^{h-(h/b)r} dr\,d\theta = \frac{h^2}{2b^2}\int_0^{2\pi}\int_0^b r(b-r)^2\,dr\,d\theta = \frac{h^2}{2b^2}\int_0^{2\pi}\left(\frac{1}{2}b^2r^2 - \frac{2b}{3}r^3 + \frac{1}{4}r^4\right)\Big]_0^b d\theta = \frac{h^2}{2b^2}\int_0^{2\pi}\left(\frac{1}{2}b^4 - \frac{2}{3}b^4 + \frac{1}{4}b^4\right) d\theta = \frac{h^2b^2}{24}\cdot 2\pi = \frac{\pi h^2 b^2}{12}$. The moment about the xy-plane is $M_{xy} = \int_0^{2\pi}\int_0^b\int_0^{h-(h/b)r} z^2 r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^b \frac{1}{3}rz^3\Big]_0^{h-(h/b)r} dr\,d\theta = \frac{h^3}{3b^3}\int_0^{2\pi}\int_0^b r(b-r)^3\,dr\,d\theta = \frac{h^3}{3b^3}\int_0^{2\pi}\left(\frac{b^3}{2}r^2 - b^2r^3 + \frac{3}{4}br^4 - \frac{1}{5}r^5\right)\Big]_0^b d\theta = \frac{h^3}{3b^3}\int_0^{2\pi}\left(\frac{b^5}{2} - b^5 + \frac{3}{4}b^5 - \frac{1}{5}b^5\right) d\theta = \frac{h^3b^2}{60}2\pi = \frac{\pi h^3 b^2}{30}$. Hence, $\bar{z} = \frac{M_{xy}}{M} = \frac{\pi h^3 b^2/30}{\pi h^2 b^2/12} = \frac{2}{5}h$. Thus, the center of mass is $\left(0, 0, \frac{2}{5}h\right)$.

44.86 Find the moments of inertia of the triangle bounded by $3x + 4y = 24$, $x = 0$, and $y = 0$, and having density 1.

▮ The moment of inertia with respect to the x-axis is $I_x = \int_0^8 \int_0^{6-(3/4)x} y^2\,dy\,dx = \int_0^8 \frac{1}{3}y^3\Big]_0^{6-(3/4)x} dx = \frac{9}{64}\int_0^8 (8-x)^3\,dx = \frac{9}{64}(-1)\frac{(8-x)^4}{4}\Big]_0^8 = \frac{9}{64}\cdot\frac{8^4}{4} = 144$. The moment of inertia with respect to the y-axis is $I_y = \int_0^8 \int_0^{6-(3/4)x} x^2\,dy\,dx = \int_0^8 x^2\cdot\frac{3}{4}(8-x)\,dx = \frac{3}{4}\left(\frac{8}{3}x^3 - \frac{x^4}{4}\right)\Big]_0^8 = \frac{3}{4}\cdot\frac{1}{12}(8)^4 = 256$.

44.87 Find the moment of inertia of a square plate of side a with respect to a side, if the density is numerically equal to the distance from an extremity of that side.

▮ Let the square be $0 \le x \le a$, $0 \le y \le a$, and let the density at (x, y) be the distance $\sqrt{x^2+y^2}$ from the origin. We want to find the moment of inertia I_x about the x-axis: $I_x = \int_0^a \int_0^a y^2\sqrt{x^2+y^2}\,dy\,dx$. Now, by the symmetry of the situation, the moment of inertia about the y-axis, $I_y = \int_0^a \int_0^a x^2\sqrt{x^2+y^2}\,dy\,dx$, must be equal to I_x. This allows us to write

$$I_x = \frac{1}{2}(I_x + I_y) = \frac{1}{2}\int_0^a \int_0^a (x^2+y^2)^{3/2}\,dy\,dx = \int_0^a \int_0^x (x^2+y^2)^{3/2}\,dy\,dx$$

where symmetry was again invoked in the last step. Change to polar coordinates and use Problem 29.9:

$$I_x = \int_0^{\pi/4}\int_0^{a\sec\theta} r^4\,dr\,d\theta = \frac{a^5}{5}\int_0^{\pi/4}\sec^5\theta\,d\theta = \frac{a^5}{5}\left[\frac{1}{4}\sec^3\theta\tan\theta + \frac{3}{8}\sec\theta\tan\theta + \frac{3}{8}\ln\tan\left(\frac{\theta}{2}+\frac{\pi}{4}\right)\right]_0^{\pi/4}$$

$$= \frac{a^5}{5}\left[\frac{\sqrt{2}}{2} + \frac{3\sqrt{2}}{8} + \frac{3}{8}\ln(1+\sqrt{2}) - 0 - 0 - 0\right] = \frac{a^5}{40}\left[7\sqrt{2} + \ln(7+5\sqrt{2})\right]$$

44.88 Find the moment of inertia of a cube of edge a with respect to an edge if the density is numerically equal to the square of the distance from one extremity of that edge.

▮ Consider the cube $0 \le x \le a$, $0 \le y \le a$, $0 \le z \le a$. Let the density be $x^2+y^2+z^2$, the square of the distance from the origin. Let us calculate the moment of inertia around the x-axis. $I_x = \int_0^a\int_0^a\int_0^a (y^2+z^2)(x^2+y^2+z^2)\,dx\,dy\,dz$. [The distance from (x, y, z) to the x-axis is $\sqrt{y^2+z^2}$.] Then $I_x = \int_0^a\int_0^a (y^2+z^2)[\frac{1}{3}x^3 + (y^2+z^2)x]\,]_0^a\,dy\,dz = \int_0^a\int_0^a (y^2+z^2)[\frac{1}{3}a^3 + (y^2+z^2)a]\,dy\,dz = \int_0^a[\frac{1}{3}a^3(\frac{1}{3}y^3+z^2y) + a(\frac{1}{5}y^5 + \frac{2}{3}y^3z^2 + z^4y)]\,]_0^a\,dz = \int_0^a[\frac{1}{3}a^3(\frac{1}{3}a^3 + az^2) + a(\frac{1}{5}a^5 + \frac{2}{3}a^3z^2 + az^4)]\,dz = a^2\int_0^a[\frac{1}{3}a^2(\frac{1}{3}a^2+z^2) + (\frac{1}{4}a^4 + \frac{2}{3}a^2z^2 + z^4)]\,dz = a^2[\frac{1}{3}a^2(\frac{1}{3}a^2z + \frac{1}{3}z^3) + (\frac{1}{4}a^4z + \frac{2}{9}a^2z^3 + \frac{1}{5}z^5)]\,]_0^a = a^2[\frac{1}{3}a^2(\frac{2}{3}a^3) + (\frac{1}{4}a^5 + \frac{2}{9}a^5 + \frac{1}{5}a^5)] = a^7(\frac{2}{9} + \frac{1}{4} + \frac{2}{9} + \frac{1}{5}) = \frac{161}{180}a^7$.

44.89 Find the centroid of the region outside the circle $r = 1$ and inside the cardioid $r = 1 + \cos\theta$.

▮ Refer to Fig. 44-20. Clearly, $\bar{y} = 0$, and $\bar{x}$ is the same for the given region as for the half lying above the polar axis. For the latter, the area $A = \int_0^{\pi/2}\int_1^{1+\cos\theta} r\,dr\,d\theta = \int_0^{\pi/2}\frac{1}{2}r^2\Big]_1^{1+\cos\theta}d\theta = \frac{1}{2}\int_0^{\pi/2}(2\cos\theta + \frac{1+\cos\theta}{2})\,d\theta = \frac{1}{2}\left(2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right)\Big]_0^{\pi/2} = \frac{1}{2}\left(2+\frac{\pi}{4}\right) = \frac{\pi+8}{8}$. The moment about the y-axis is $M_y = \int_0^{\pi/2}\int_1^{1+\cos\theta} xr\,dr\,d\theta = \int_0^{\pi/2}\int_1^{1+\cos\theta} r^2\cos\theta\,dr\,d\theta = \int_0^{\pi/2}\frac{1}{3}r^3\cos\theta\Big]_1^{1+\cos\theta}d\theta = \frac{1}{3}\int_0^{\pi/2}[(1+\cos\theta)^3 - 1]\cos\theta\,d\theta = \frac{1}{3}\int_0^{\pi/2}(3\cos^2\theta + 3\cos^3\theta + \cos^4\theta)\,d\theta = \frac{1}{3}\int_0^{\pi/2}\left[\frac{3}{2}(1+\cos 2\theta) + 3(\cos\theta - \sin^2\theta\cos\theta) + \frac{1}{4}\left(1 + 2\cos 2\theta + \frac{1+\cos 4\theta}{2}\right)\right]\Big]_0^{\pi/2} = \frac{1}{3}\left(\frac{15}{8}\theta + 3\sin\theta + \sin 2\theta + \frac{\sin 4\theta}{32} - \sin^3\theta\right)\Big]_0^{\pi/2} = \frac{1}{3}\left(\frac{15\pi}{16} + 3 - 1\right) = \frac{5\pi}{16} + \frac{2}{3} = \frac{15\pi+32}{48}$. Hence, $\bar{x} = \frac{M_y}{A} = \frac{(15\pi+32)/48}{(\pi+8)/8} = \frac{15\pi+32}{6(\pi+8)}$.

44.90 Find the centroid $(\bar{x}, \bar{y})$ of the region in the first quadrant bounded by $y^2 = 6x$, $y = 0$, and $x = 6$ (Fig. 44-43).

▮ The area $A = \int_0^6\int_{y^2/6}^6 dx\,dy = \int_0^6\left(6 - \frac{y^2}{6}\right)dy = \left(6y - \frac{1}{18}y^3\right)\Big]_0^6 = 6(6-2) = 24$. The moment about the x-axis is $M_x = \int_0^6\int_{y^2/6}^6 y\,dx\,dy = \int_0^6 y\left(6 - \frac{y^2}{6}\right)dy = \left(3y^2 - \frac{1}{24}y^4\right)\Big]_0^6 = 36\left(3 - \frac{3}{2}\right) = 54$. Hence, $\bar{y} = \frac{M_x}{A} = \frac{54}{24} = \frac{9}{4}$. The moment about the y-axis is $M_y = \int_0^6\int_{y^2/6}^6 x\,dx\,dy = \int_0^6\frac{1}{2}x^2\Big]_{y^2/6}^6 dy = \frac{1}{2}\int_0^6\left(36 - \frac{y^4}{36}\right)dy = \frac{1}{2}\left(36y - \frac{1}{180}y^5\right)\Big]_0^6 = 3\left(36 - \frac{36}{5}\right) = \frac{12\cdot 36}{5}$. Hence, $\bar{x} = \frac{M_y}{A} = \frac{12(36)/5}{24} = \frac{18}{5}$. So the centroid is $\left(\frac{18}{5}, \frac{9}{4}\right)$.

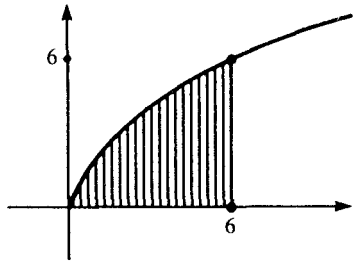


Fig. 44-43

44.91 Find the centroid of the solid under $z^2 = xy$ and above the triangle bounded by $y = x$, $y = 0$, and $x = 4$.

▮ The volume $V = \int_0^4\int_0^x\int_0^{\sqrt{xy}} dz\,dy\,dx = \int_0^4\int_0^x\sqrt{xy}\,dy\,dx = \int_0^4\frac{2}{3}x^{1/2}y^{3/2}\Big]_0^x dx = \int_0^4\frac{2}{3}x^2\,dx = \frac{2}{9}x^3\Big]_0^4 = \frac{128}{9}$. The moment about the yz-plane is $M_{yz} = \int_0^4\int_0^x\int_0^{\sqrt{xy}} x\,dz\,dy\,dx = \int_0^4\int_0^x xz\,]_0^{\sqrt{xy}}\,dy\,dx = \int_0^4\int_0^x x^{3/2}y^{1/2}\,dy\,dx = \int_0^4\frac{2}{3}x^{3/2}y^{3/2}\Big]_0^x dx = \int_0^4\frac{2}{3}x^3\,dx = \frac{1}{6}x^4\Big]_0^4 = \frac{128}{3}$. Hence, $\bar{x} = \frac{M_{yz}}{V} = \frac{\frac{128}{3}}{\frac{128}{9}} = 3$. The moment about the

xz-plane is $M_{xz} = \int_0^4 \int_0^x \int_0^{\sqrt{xy}} y\,dz\,dy\,dx = \int_0^4 \int_0^x x^{1/2}y^{3/2}\,dy\,dx = \int_0^4 \frac{2}{5}x^{1/2}y^{5/2}\Big]_0^x dx = \int_0^4 \frac{2}{5}x^3\,dx = \frac{1}{10}x^4\Big]_0^4 = \frac{128}{5}$. Hence, $\bar{y} = \frac{M_{xz}}{V} = \frac{\frac{128}{5}}{\frac{128}{9}} = \frac{9}{5}$. The moment about the xy-plane is $M_{xy} = \int_0^4 \int_0^x \int_0^{\sqrt{xy}} z\,dz\,dy\,dx = \int_0^4 \int_0^x \frac{1}{2}z^2\Big]_0^{\sqrt{xy}} dy\,dx = \int_0^4 \int_0^x \frac{1}{2}xy\,dy\,dx = \int_0^4 \frac{1}{4}xy^2\Big]_0^x dx = \int_0^4 \frac{1}{4}x^3\,dx = \frac{1}{16}x^4\Big]_0^4 = 16$. Hence, $\bar{z} = \frac{M_{xy}}{V} = \frac{16}{\frac{128}{9}} = \frac{9}{8}$. Thus, the centroid is $\left(3, \frac{9}{5}, \frac{9}{8}\right)$.

44.92 Find the centroid of the upper half $\mathscr{H}$ of the solid ball of radius a with center at the origin.

▮ We know $V = \frac{1}{2}(\frac{4}{3}\pi a^3) = \frac{2}{3}\pi a^3$. By symmetry, $\bar{x} = \bar{y} = 0$. The moment about the xy-plane is $M_{xy} = \iiint_{\mathscr{H}} z\,dV$. Use spherical coordinates.

$$M_{xy} = \int_0^{2\pi} \int_0^{\pi/2} \int_0^a \rho\cos\phi \cdot \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta = \int_0^{2\pi} \int_0^{\pi/2} \tfrac{1}{4}\rho^4 \cos\phi\sin\phi\,]_0^a\,d\phi\,d\theta$$

$$= \frac{a^4}{4}\int_0^{2\pi}\int_0^{\pi/2} \cos\phi\sin\phi\,d\phi\,d\theta = \frac{a^4}{4}\int_0^{2\pi} \frac{1}{2}\sin^2\phi\Big]_0^{\pi/2} d\theta = \frac{a^4}{8}\int_0^{2\pi} d\theta = \frac{a^4}{8}\cdot 2\pi = \frac{\pi a^4}{4}$$

Hence, $\bar{z} = \frac{M_{xy}}{V} = \frac{\pi a^4/4}{2\pi a^3/3} = \frac{3}{8}a$.